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Analytics & Finance Professional

Power BI · Python · Power Automate · SQL · PowerShell · Big 4 Audit Foundation · SOX Compliance

1,000+	10+	\$300K+	\$250K+
Annual hours automated	Enterprise dashboards deployed	NOI impact potential	Deposit recovery (first weeks)

Every project in this portfolio is production work that is actively used by executive, operational, and field teams across a publicly traded national REIT.

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1. Third-Party Utility Billing - Dashboard & Automation

~53 hrs	~630 hrs	300+	\$300K+
Saved monthly	Saved annually	Reports auto-pulled	NOI impact potential

THE PROBLEM

Utility expenses recovered outside of CAM through a third-party billing provider had low recovery rates, accounting errors, aged AR, and a lack of visibility. Analyzing the expenses was time-intensive requiring manually downloading and comparing individual PDF reports from separate regional emails. Portfolio-wide monitoring and analysis was extremely difficult.

THE SOLUTION

- Built a Power Automate Desktop RPA + PowerShell scripting to automatically pull and consolidate 300+ reports from the vendor website bi-monthly with zero manual downloads
- Python scripts (pdfplumber + pandas) parse and consolidate unstructured text PDF invoice data into structured tables for Power BI
- Power BI delivers AR, billing, consumption, and GL allocation views at National, Division, Regional, and Property level which is visibility that did not exist before
- Automated daily email distribution of utility invoices to Property Management teams by Regional Property Manager

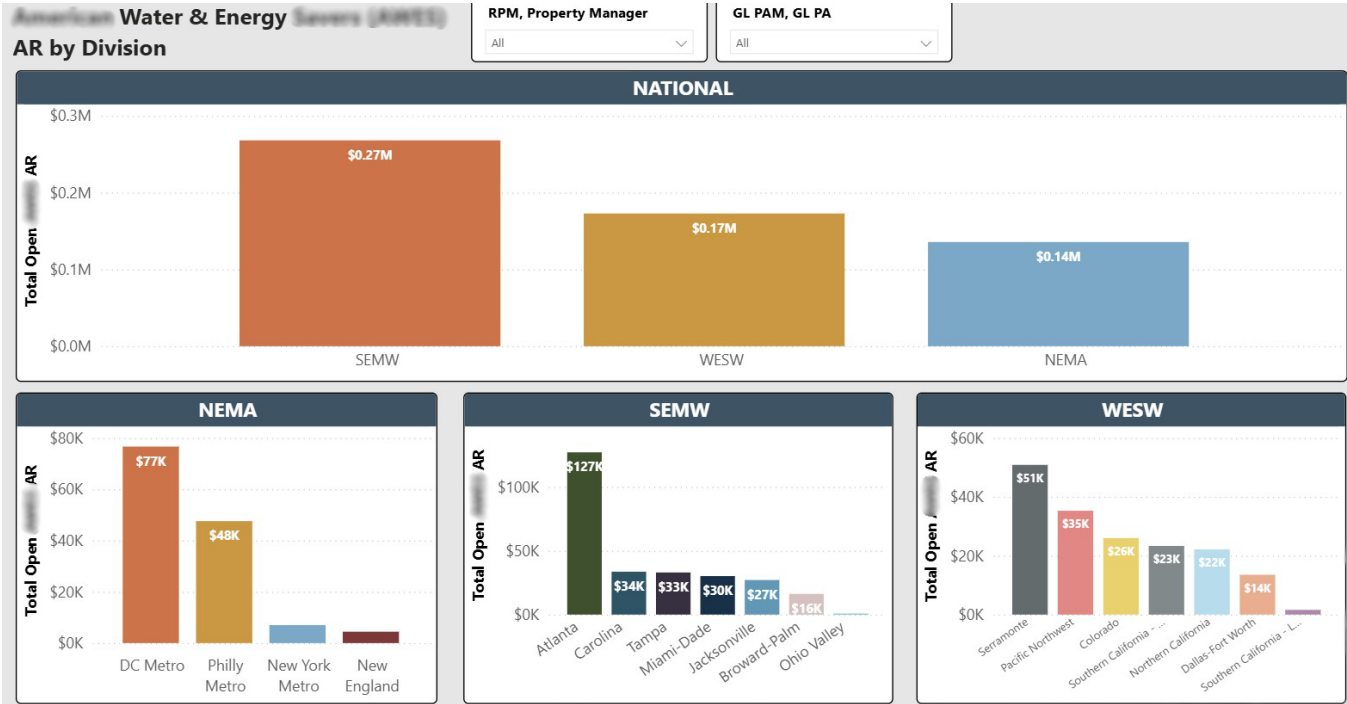
THE RESULT

- 53 hours saved monthly across Property Accounting, Property Management, and Property Operations (630 hours annually)
- \$300K+ increase in NOI-related utility recoveries since rollout which has been driven by visibility into aged AR over 120 days and broken meter identification utilizing estimated usage for billbacks
- Regional leaders can now identify non-paying tenants and broken meters across their portfolio in seconds
- Note: Dollar figures shown in dashboard views are filtered to a subset for portfolio anonymization.

The automation + Power BI combination was the first portfolio-wide utility analytics solution at the company. Previously there was no way to view AR, consumption, or billing trends across all 106 properties simultaneously.

Dashboard View 1: AR by Division & Region

National view showing open AR broken out by division (SEMW, WESW, NEMA) and drilled down by region. Colored bars allow rapid identification of high-balance areas. Filterable by Division, Region, RPM, PM, and GL PAM.



AR by Division & Region: National overview with SEMW, WESW, and NEMA breakdowns

Dashboard View 2: AR Detail by Tenant

Property and tenant-level AR aging table showing 0–30, 31–60, 61–90, 91–120, and 121+ day buckets. Includes estimated vs actual meter read indicators and last payment dates.

American Water & Energy Services (AWES) - Tenant AR (UT_FIN102)												
Division, Region	Shopping Center	RPM, Property Manager	GL PAM, GL PA	Estimated Usage (Y/N)?	Show Zero Balances?							
All	All	All	All	All	All							
Shopping Center	BU Name	Resident	0-30 days	31-60 days	61-90 days	91-120 days	121+ days	Balance	Last Paid Date	Due Date	Estimated Usage (Y/N)?	
Shopping Center 001	BU 001	Resident 001	\$130.65	\$135.84	\$134.48	\$64.79	\$28,316.21	\$28,781.97	10/1/2019	4/25/2026	Y	
Shopping Center 002	BU 002	Resident 002	\$593.38	\$1,061.87	\$1,844.83	\$2,229.42	\$12,817.37	\$18,546.87	2/3/2026	4/25/2026	N	
Shopping Center 003	BU 003	Resident 003	\$416.28	\$442.58	\$416.28	\$442.58	\$14,916.74	\$16,634.46	4/20/2022	4/25/2026	Y	
Shopping Center 004	BU 004	Resident 004	\$0.00	\$5,723.05	\$0.00	\$0.00	\$6,012.90	\$11,735.95	10/29/2025	5/5/2026	N	
Shopping Center 005	BU 005	Resident 005	\$358.52	\$379.63	\$358.52	\$379.63	\$9,439.13	\$10,915.43		4/25/2026	Y	
Shopping Center 006	BU 006	Resident 006	\$302.26	\$319.35	\$302.26	\$319.35	\$8,924.97	\$10,168.19	9/20/2021	4/25/2026	Y	
Shopping Center 007	BU 007	Resident 007	\$0.00	\$2,862.64	\$0.00	\$0.00	\$6,760.40	\$9,623.04		5/5/2026	N	
Shopping Center 008	BU 008	Resident 008	\$296.14	\$296.14	\$296.14	\$296.14	\$8,288.26	\$9,472.82	4/24/2023	4/25/2026	Y	
Shopping Center 009	BU 009	Resident 009	\$125.39	\$148.55	\$86.71	\$85.65	\$8,936.79	\$9,383.09	4/26/2026	4/25/2026	N	
Shopping Center 010	BU 010	Resident 010	\$222.74	\$254.05	\$274.62	\$319.77	\$6,995.72	\$8,066.90	9/19/2025	4/25/2026	N	
Shopping Center 011	BU 011	Resident 011	\$749.40	\$660.99	\$615.73	\$759.22	\$4,596.89	\$7,382.23		4/25/2026	N	
Shopping Center 012	BU 012	Resident 012	\$729.29	\$816.76	\$672.70	\$749.87	\$4,324.77	\$7,293.39		4/25/2026	N	
Shopping Center 013	BU 013	Resident 013	\$0.00	\$3,570.53	\$0.00	\$0.00	\$2,958.61	\$6,529.14	11/19/2025	5/5/2026	N	
Shopping Center 014	BU 014	Resident 014	\$0.00	\$0.00	\$0.00	\$0.00	\$6,371.57	\$6,371.57	3/4/2022	11/25/2025	N	
Shopping Center 015	BU 015	Resident 015	\$555.62	\$579.29	\$589.58	\$681.15	\$3,914.78	\$6,320.42		4/25/2026	N	
Shopping Center 016	BU 016	Resident 016	\$574.20	\$615.36	\$543.33	\$625.65	\$3,664.53	\$6,023.07		4/25/2026	N	
Shopping Center 017	BU 017	Resident 017	\$132.05	\$194.27	\$98.45	\$103.10	\$5,190.57	\$5,718.44	12/5/2024	4/25/2026	N	
Shopping Center 018	BU 018	Resident 018	\$229.51	\$241.63	\$229.51	\$241.63	\$4,746.25	\$5,688.53	2/16/2024	4/25/2026	Y	
Total			\$2,299.51	\$2,416.33	\$2,299.51	\$2,416.33	\$4,746.25	\$5,688.53				

AR Detail: Tenant-level aging with estimated vs actual meter read flags; filtered subset shown

Dashboard View 3: Billed Consumption

Consumption tracking by property, meter, and billing cycle. Allows Property Management to identify properties with unusual consumption patterns or estimated reader reads which is a key driver of the NOI improvement post-rollout.

American Water & Energy Services (AWES) - Billed Consumption (UT_CON116)											
Division, Region		Shopping Center		RPM, Property Manager		GL PAM, GL PA		Estimated Usage (Y/N)?			
All		All		All		All		All			
BU #	BU Name	Building	Premise #	Occupied By	Utility	Prior Read Date	Current Read Date	Meter #	Consumption	Estimated Usage (Y/N)?	
1000	1000	1000	A	1000	Water Rate	02/26/26	03/27/26	A	5,580.08	N	
1000	1000	1000	B-C	1000	Water Rate	02/26/26	03/27/26	B-C	9,800.00	N	
1000	1000	1000	D	1000	Water Rate	02/26/26	03/27/26	D	7,659.52	N	
1000	1000	1000	18-19	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266017	3,500.00	N	
1000	1000	1000	18A	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266016	0.00	Y	
1000	1000	1000	16	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266015	18,630.00	Y	
1000	1000	1000	15	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266014	8,710.00	Y	
1000	1000	1000	14	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266013	5,100.00	N	
1000	1000	1000	12	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266011	5,780.00	N	
1000	1000	1000	11 CTU	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266010	146.80	N	
1000	1000	1000	8	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266008	440.00	N	
1000	1000	1000	9	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266009	340.00	Y	
1000	1000	1000	7	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266007	400.00	N	
1000	1000	1000	6 CTU	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266006	30,990.30	N	
1000	1000	1000	5 CTU	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266005	0.00	N	
1000	1000	1000	4	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266004	13,330.00	N	
1000	1000	1000	13 CTU	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266012	6,279.80	N	
Total			2 CTU	1000	Sewer Rate, Water Rate	03/29/26	04/27/26	19266002	0.00	N	

Billed Consumption: Property and meter-level usage tracking with estimated read indicators

Dashboard View 4: Invoice Totals by Property (Python-Generated)

Python script consolidates individual unstructured text PDF invoices from the vendor into a structured excel table which this Power BI report pulls as a source. This view shows total invoice amounts by property and billing cycle — the output that feeds the AR and billing Power BI views. Totals shown are a filtered subset to anonymize.

American Water & Energy Services (AWES) - Simple Invoice (BIL111) by Total Invoice Amount by Property						
Division, Region	Shopping Center	RPM, Property Manager	GL PAM, GL PA	Bill Cycle		
All	All	All	All	All		
BusinessUnitFullName	BU Name	Bill Cycle	TYPE	Date From	Date To	Total Invoice Amount
Commercial - 2025	Commercial - 2025	Commercial	submetered	10/29/2025	1/28/2026	\$46,420.35
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$44,355.27
City Cycle - 2025	City Cycle - 2025	City Cycle	submetered	12/10/2025	3/9/2026	\$15,420.51
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$12,240.35
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$11,611.05
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$9,388.19
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$8,612.53
City Cycle - 2026	City Cycle - 2026	City Cycle	submetered	4/9/2026	5/7/2026	\$8,508.87
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$8,156.45
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$7,750.10
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$7,546.70
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$7,360.71
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$7,327.26
TABS - 2026	TABS - 2026	TABS	TABS	1/7/2026	3/4/2026	\$7,170.20
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$6,837.61
TABS - 2026	TABS - 2026	TABS	TABS	1/24/2026	3/23/2026	\$6,708.82
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$6,117.78
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$5,720.76
City Cycle - 2025	City Cycle - 2025	City Cycle	submetered	8/28/2025	9/19/2025	\$5,482.27
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$5,434.35
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$5,264.37
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$5,213.11
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$5,079.36
Commercial - 2026	Commercial - 2026	Commercial	submetered	3/30/2026	4/27/2026	\$4,558.14
Total		Commercial	submetered	3/30/2026	4/27/2026	\$4,409.87

Invoice Totals by Property: Python-generated consolidation of 100+ individual PDF invoices

Dashboard View 5: Engie Invoices by Property & Service Type

Meter-level invoice detail showing service type (Water, Sewer, Electric, Gas), account numbers, service dates, and individual invoice amounts. Includes a direct hyperlink to each invoice image for immediate drill-through to source documentation.

American Water & Energy Services (AWES) - Engie Invoices by Property & Service Type											
Division, Region		Shopping Center		RPM, Property Manager		GL PAM, GL PA		CUSTOMER GL #		BILL DATE	
All		All		All		All		All		All	
LOCATION #	LOCATION NAME	SERVICE TYPE	ACCOUNT #	METER #	SERVICE BEGIN DATE	SERVICE END DATE	BILL DATE	CUSTOMER GL #	Unit of Measure	Total Cost	Utility Invoice (Click for Invoice)
		Sewer	0000309165488	NDO8	9/30/2025	10/30/2025	11/3/2025	7643	kGal	560.77	https://layouts/ExternalImages/BillImage.aspx?s=jPp08fpAVWx6WKsNaxv1kbbjvxlX8gHRUHVRLc7WRVY%3d
		Water	0000309165488	NDO8	9/30/2025	10/30/2025	11/3/2025	7643	kGal	182.30	https://layouts/ExternalImages/BillImage.aspx?s=jPp08fpAVWx6WKsNaxv1kbbjvxlX8gHRUHVRLc7WRVY%3d
		Water	2152340000	416411	10/2/2025	11/1/2025	11/3/2025	7643	kGal	75.06	https://layouts/ExternalImages/BillImage.aspx?s=fp6kimDbprzEPrrFiliuY8Zl7j0P61E2G6wiS4ZNUA%3d
		Water	9052340000	443933	10/2/2025	11/1/2025	11/3/2025	7643	kGal	369.84	https://layouts/ExternalImages/BillImage.aspx?s=mj2ewci3dTPpodbT9iFG4eIMn9IXSk5mVzeTOdmHss%3d
		Sewer	1399300000	MM	10/4/2025	11/3/2025	11/4/2025	7643	kGal	6,704.25	https://layouts/ExternalImages/BillImage.aspx?s=i68KnZyWLaOt8CefftdxCukLGDESE5wu8XGHIRbWlqY%3d
		Water	1399300000	702997700	10/4/2025	11/3/2025	11/4/2025	7643	kGal	0.00	https://layouts/ExternalImages/BillImage.aspx?s=i68KnZyWLaOt8CefftdxCukLGDESE5wu8XGHIRbWlqY%3d
		Water	1399300000	703000048	10/4/2025	11/3/2025	11/4/2025	7643	kGal	0.00	https://layouts/ExternalImages/BillImage.aspx?s=i68KnZyWLaOt8CefftdxCukLGDESE5wu8XGHIRbWlqY%3d
		Water	1399300000	72227558H	10/4/2025	11/3/2025	11/4/2025	7643	kGal	1,587.86	https://layouts/ExternalImages/BillImage.aspx?s=i68KnZyWLaOt8CefftdxCukLGDESE5wu8XGHIRbWlqY%3d
		Water	1399300000	MM	10/4/2025	11/3/2025	11/4/2025	7643	kGal	1,962.10	https://layouts/ExternalImages/BillImage.aspx?s=i68KnZyWLaOt8CefftdxCukLGDESE5wu8XGHIRbWlqY%3d
Total											

Engie Invoices by Property & Service Type: Meter-level detail with clickable invoice document links

Dashboard View 6: GL Allocations

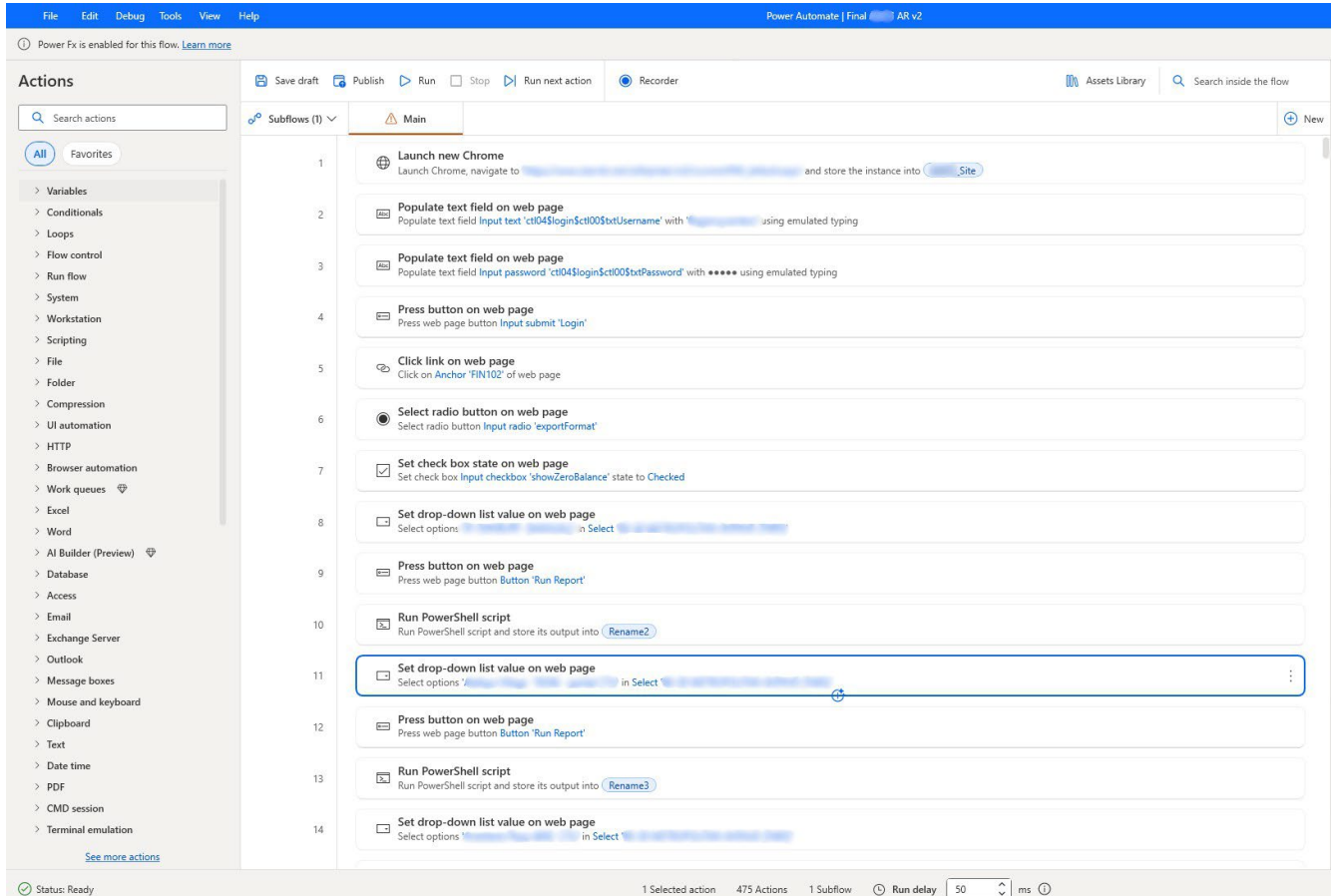
GL allocation mapping showing how each utility meter's charges map to specific GL accounts, cost centers, and allocation percentages. Used by Property Accounting to verify journal entry coding and identify misallocated charges.

American Water & Energy Services (AWES) - Engie GL Allocations										
Division, Region	Shopping Center	RPM, Property Manager	GL PAM, GL PA	CUSTOMER GL #	SERVICE TYPE	SERVICE STATUS				
All	All	All	All	All	All	All				
LOCATION #	LOCATION NAME	ACCOUNT #	METER #	SERVICE TYPE	VENDOR NAME	ACCOUNT STATUS	SERVICE STATUS	CUSTOMER GL #	GL DESCRIPTION	GL % ALLOCATION
		0 5000 5989 5914		Late Charges	Gas	Inactive	Inactive	7659.1	NON RECOV-UTIL LATE FEES	100
		0 5000 5989 5914		Misc Charges	Gas	Inactive	Inactive	4480.2	TENANT REIMBURSABLE EXPE	100
		0 5000 5989 5914		Natural Gas	Gas	Inactive	Inactive	4480.2	TENANT REIMBURSABLE EXPE	100
		0 5000 5989 5914		Other Services	Gas	Inactive	Inactive	1255	Deposit Refund	100
		0 5000 5989 5914		Tax	Gas	Inactive	Inactive	4480.2	TENANT REIMBURSABLE EXPE	100
		0 5000 5989 5914		Natural Gas	Gas	Inactive	Inactive	4480.2	TENANT REIMBURSABLE EXPE	100
		0000 92592-566822 9		Electric	General Electric	Inactive	Inactive	7655.1	VACANCY	100
		0000 92592-566822 9		Late Charges	General Electric	Inactive	Inactive	7659.1	NON RECOV-UTIL LATE FEES	100
		0000 92592-566822 9		Misc Charges	General Electric	Inactive	Inactive	7655.1	VACANCY	100
		0000 92592-566822 9		Other Services	General Electric	Inactive	Inactive	7655.1	VACANCY	100
		0000 92592-566822 9		Tax	General Electric	Inactive	Inactive	7655.1	VACANCY	100
		0000 92592-566822 9		Electric	General Electric	Inactive	Inactive	7655.1	VACANCY	100
		0000 92592-566822 9		Late Charges	General Electric	Inactive	Inactive	7655.1	VACANCY	100
		0000 92592-566822 9		Misc Charges	General Electric	Inactive	Inactive	7659.1	NON RECOV-UTIL LATE FEES	100
		0000 97410-566553 9		Tax	General Electric	Inactive	Inactive	7655.1	VACANCY	100
		0000 97410-566553 9		Electric	General Electric	Inactive	Inactive	7210	ELECTRIC	100
		0000 97410-607518 3		Late Charges	General Electric	Inactive	Inactive	7659.1	NON RECOV-UTIL LATE FEES	100
		0000 97410-607518 3		Misc Charges	General Electric	Inactive	Inactive	7210	ELECTRIC	100
		0000 97410-607518 3		Tax	General Electric	Inactive	Inactive	7210	ELECTRIC	100
		0000 97410-607518 3		Electric	General Electric	Inactive	Inactive	7655.1	VACANCY COSTS -UTILITIES	100
		0000 97410-619743 3		Late Charges	General Electric	Inactive	Inactive	7659.1	NON RECOV-UTIL LATE FEES	100
		0000 97410-619743 3		Misc Charges	General Electric	Inactive	Inactive	7655.1	VACANCY COSTS -UTILITIES	100
		0000 97410-619743 3		Tax	General Electric	Inactive	Inactive	7655.1	VACANCY	100
		0000 97410-619743 3		Electric	General Electric	Inactive	Inactive	7655.1	VACANCY COSTS -UTILITIES	100
		0000 97410-619744 1		Late Charges	General Electric	Inactive	Inactive	7659.1	NON RECOV-UTIL LATE FEES	100
		0000 97410-619744 1		Misc Charges	General Electric	Inactive	Inactive	7655.1	VACANCY COSTS -UTILITIES	100
		0000 97410-619744 1		Other Services	General Electric	Inactive	Inactive	7655.1	VACANCY COSTS -UTILITIES	100

GL Allocations: Account, meter, service type, vendor, and GL description with allocation percentage per line

Automation: Power Automate Desktop Flow

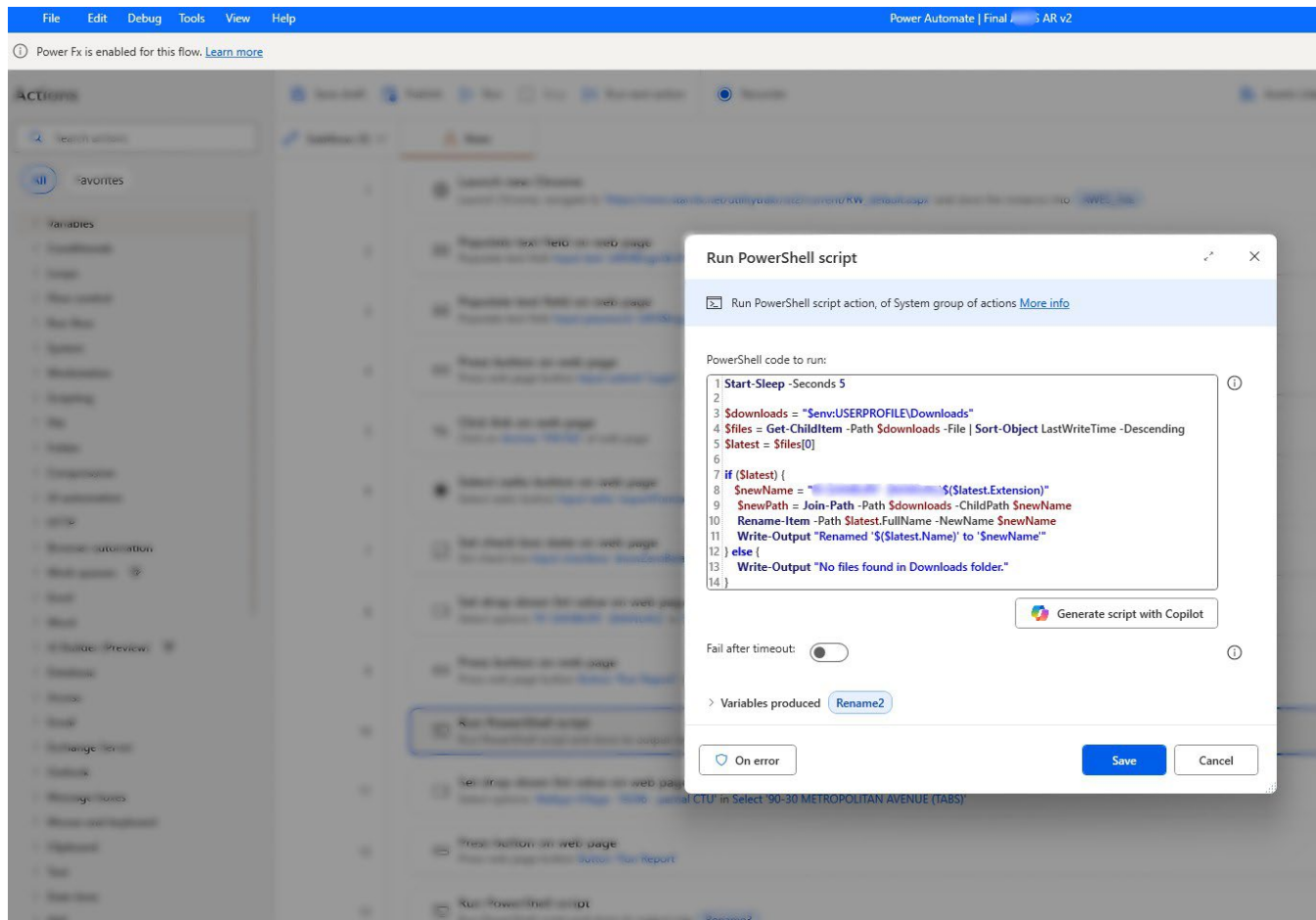
The RPA flow runs on a scheduled basis and orchestrates the full automation cycle: log into the vendor portal, navigate to report download pages, iterate through each property's reports, download and save files, then trigger the PowerShell processing scripts.



Power Automate Desktop RPA flow: automated report extraction from vendor website

Automation: PowerShell (Snip 1)

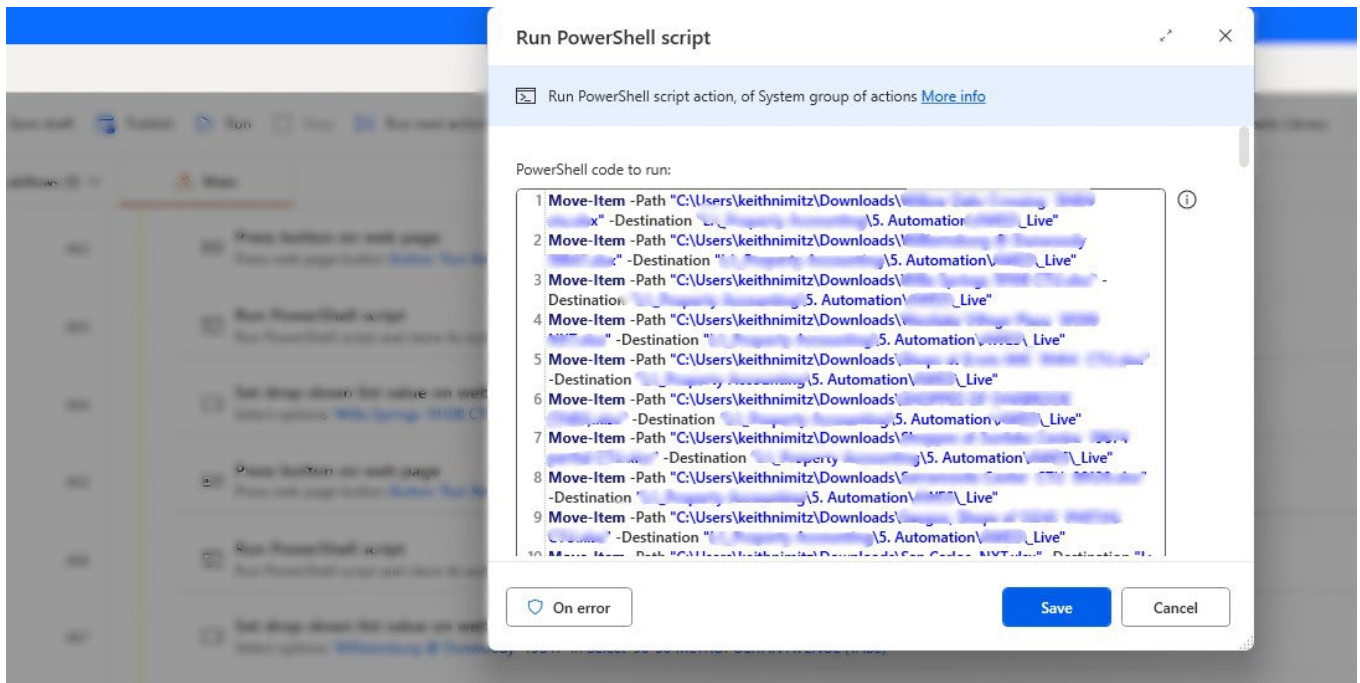
PowerShell scripts handle file routing, folder organization, and triggering subsequent automation steps. Includes error handling, logging, and automatic notification if a file cannot be processed.



PowerShell: file handling, routing logic, and error notification (Snip 1 of 2)

Automation: PowerShell (Snip 2)

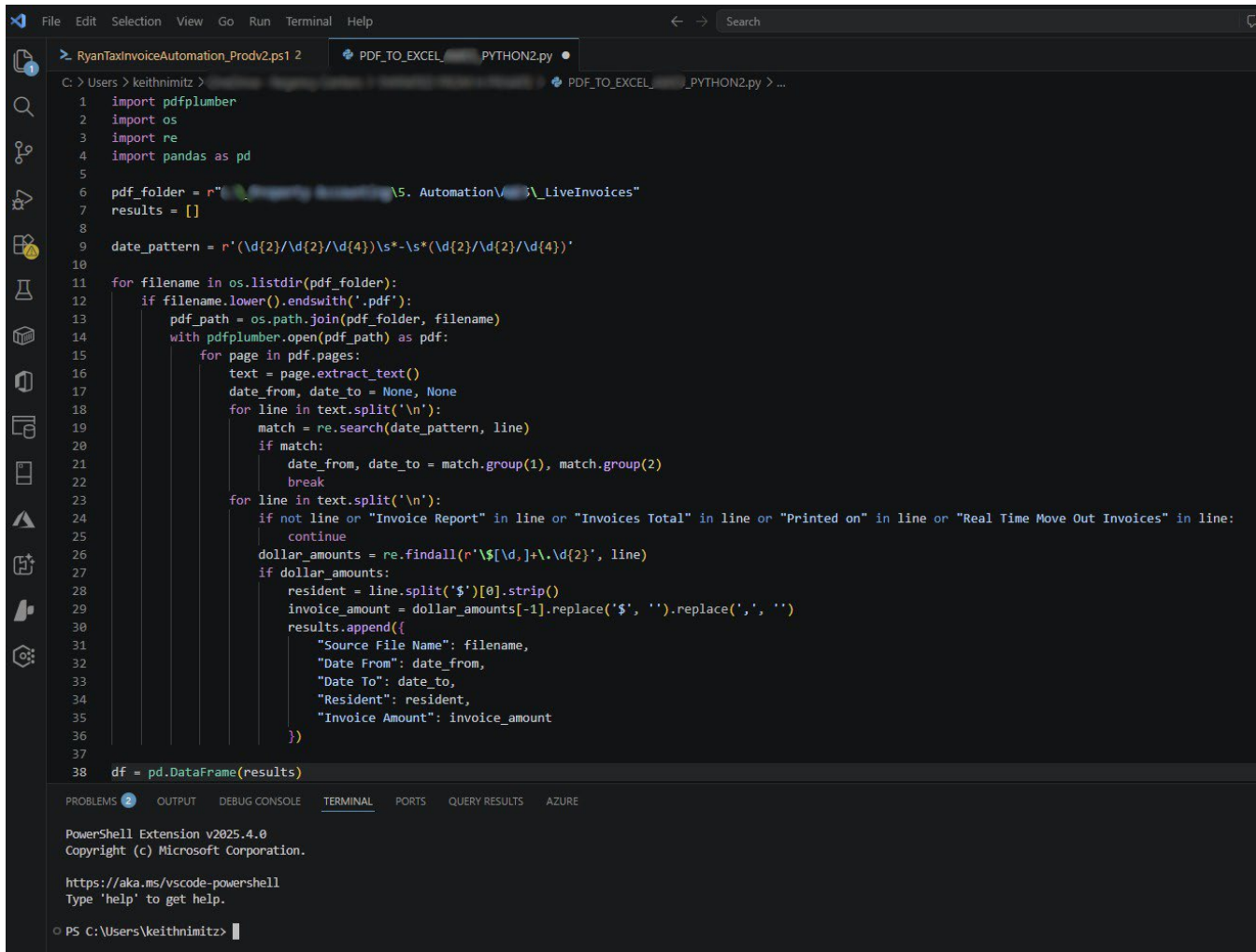
Continuation of the PowerShell routing and logging logic showing additional processing steps and error handling branches.



PowerShell: continuation with additional routing and logging logic (Snip 2 of 2)

Python: PDF Parsing & Data Extraction

Python script using pdfplumber and pandas libraries to open each PDF invoice, extract data fields from the unstructured text (dates, amounts, property identifiers), and consolidate into a single DataFrame for Power BI consumption. Regular expressions handle date range and dollar amount parsing.



```
File Edit Selection View Go Run Terminal Help
RyanTaxInvoiceAutomation_Prodv2.ps1 2 PDF_TO_EXCEL PYTHON2.py
C:\Users\keithnimitz>
1 import pdfplumber
2 import os
3 import re
4 import pandas as pd
5
6 pdf_folder = r"C:\Users\keithnimitz\Documents\Automation\LiveInvoices"
7 results = []
8
9 date_pattern = r'(\d{2}/\d{2}/\d{4})\s*-\s*(\d{2}/\d{2}/\d{4})'
10
11 for filename in os.listdir(pdf_folder):
12     if filename.lower().endswith('.pdf'):
13         pdf_path = os.path.join(pdf_folder, filename)
14         with pdfplumber.open(pdf_path) as pdf:
15             for page in pdf.pages:
16                 text = page.extract_text()
17                 date_from, date_to = None, None
18                 for line in text.split('\n'):
19                     match = re.search(date_pattern, line)
20                     if match:
21                         date_from, date_to = match.group(1), match.group(2)
22                     break
23                 for line in text.split('\n'):
24                     if not line or "Invoice Report" in line or "Invoices Total" in line or "Printed on" in line or "Real Time Move Out Invoices" in line:
25                         continue
26                     dollar_amounts = re.findall(r'\$\d{,}[.]\d{2}', line)
27                     if dollar_amounts:
28                         resident = line.split('$')[0].strip()
29                         invoice_amount = dollar_amounts[-1].replace('$', '').replace(',', '')
30                         results.append({
31                             "Source File Name": filename,
32                             "Date From": date_from,
33                             "Date To": date_to,
34                             "Resident": resident,
35                             "Invoice Amount": invoice_amount
36                         })
37
38 df = pd.DataFrame(results)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS AZURE

PowerShell Extension v2025.4.0
Copyright (c) Microsoft Corporation.

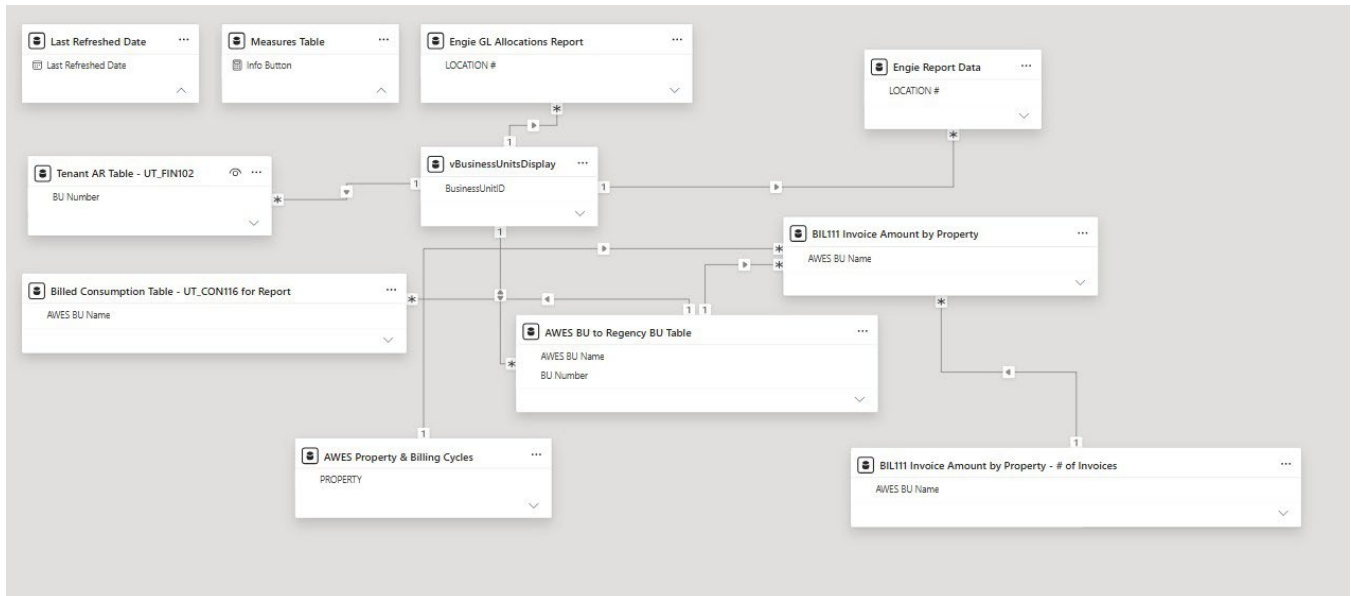
<https://aka.ms/vscode-powershell>
Type 'help' to get help.

PS C:\Users\keithnimitz>

Python (pdfplumber + pandas + re) — PDF invoice parsing and structured data extraction

Data Model: Semantic Model Architecture

9-table semantic model bridging AWES third-party provider data to internal business unit structure. Key design: a custom AWES BU to Regency BU mapping table bridges the external vendor's property naming to the company's internal BU hierarchy, enabling consistent filtering and drill-down across all views.



Semantic model: 9 tables connecting AWES data, Engie reports, tenant AR, consumption, GL allocations, and BU mapping

2. Tenant Dispute Database - Dashboards & Automation

~290 hrs	48+ hrs	Daily	2 views
Saved annually (reporting)	Saved annually (email follow-ups)	Dashboard refresh	Leadership + Operational

THE PROBLEM

Reporting on tenant disputes was time-intensive requiring 3 days from the Tenant Disputes Manager to prepare monthly and quarterly. Monitoring hundreds of open disputes across multiple workflow stages required constant manual follow-up. These challenges contributed to longer turnaround times on dispute resolution.

THE SOLUTION

- Coordinated with the dispute system vendor and IT teams to access the system API and build nightly data loads into the company data warehouse
- Validated data warehouse tables directly using SQL queries against ADF.TenantDispute and ADF.vTenantDisputeTransactionHistory verifying data integrity before building the reporting layer
- Built two Power BI reports: Sr. Leadership view (KPI cards, drillthrough, trend analysis) and Operational view (workflow status, aging, comments)
- Both reports refresh automatically daily from data warehouse tables loaded nightly from the API
- Power Automate Cloud flows query the Power BI semantic model and auto-send escalating reminders: 48 hours, 14 days, 30 days (manager CC'd), then weekly until resolved

THE RESULT

- Dispute reporting reduced from 3 days of manual preparation to on-demand real-time dashboard
- Saves the Tenant Disputes Manager 24 hours monthly (290 hours annually)
- Automated email escalations save an additional ~48 hours annually in manual follow-up
- Leadership has timely visibility into open dispute volumes, aging, and trends — without manual report preparation
- Note: Dollar amounts and dispute counts shown in views are filtered subsets for confidentiality.

Key differentiator: This project required working across three systems (the dispute vendor's API, the company's IT/data warehouse team, and Power BI) and coordinating a full data pipeline before a single report could be built. The automation layer makes it a complete end-to-end solution.

Sr. Leadership Dashboard: Divisional View

Summary page showing open dispute KPI cards (count, aging, amounts) with a hierarchy matrix drillable from Division → Regional Officer → Property Manager → Shopping Center → Tenant. Toggle between Regional Officer View and Tenant View using the navigation buttons.

Tips for this Dashboard:
How do I quickly filter on the disputes related to my Division, Portfolio, etc.?

- To filter the card values to the right by Division, Regional Officer, or other descriptor, select the value in the matrix table below. For example, if I wanted the card values to adjust for the Northeast and MidAtlantic I would click "Northeast and MidAtlantic" in the table below. The selection would then highlight gray and update the card values.

How do I find more detail on what makes up the values?

- For a more detailed breakdown on any value, you can "drillthrough" by right-clicking the value in question and selecting the relevant drillthrough for the additional details you are interested in. From there you will be directed to a table with a breakdown of what makes up the value.

How do I view Open Dispute Trends?

- By clicking on the three bars or "hamburger" at the top right of the page, you will see additional pages available for you including an Opened Disputes graph trended over time.

Regional Officer View

Tenant View

of Disputes Opened in 2025
1K

of Disputes Opened in 2026
333

of Disputes Open > \$20k
27

of Disputes Open > 180 Days
24

of Open Disputes
147

Avg Days Open Disputes > \$20k
132

Avg Days Open Disputes > 180 Days
242

Avg Days Open
64

Avg Amount Disputed > \$20k
\$38.16K

Avg Amount Disputed > 180 Days
\$16.71K

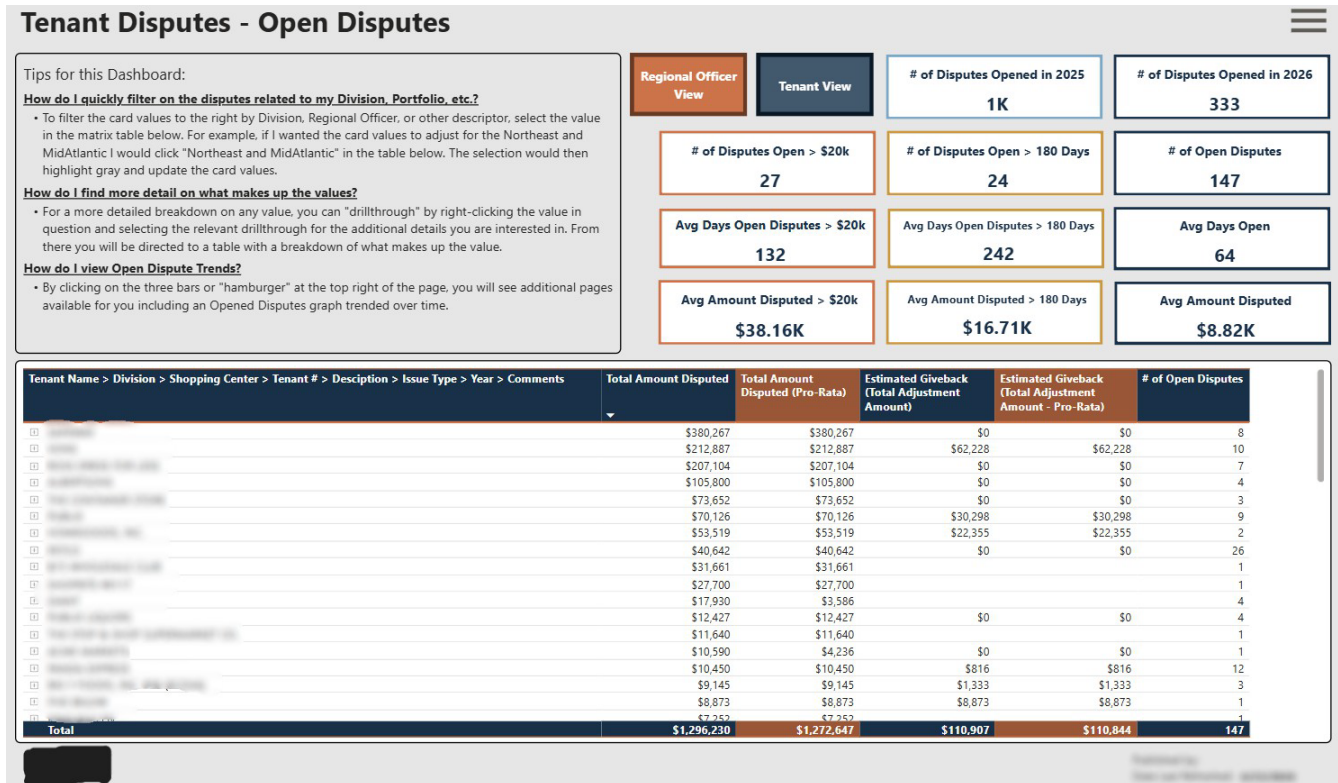
Avg Amount Disputed
\$8.82K

Division > Regional Officer > Regional Prop Manager > Shopping Center > Tenant Info > Dispute Info	Total Amount Disputed	Total Amount Disputed (Pro-Rata)	Estimated Giveback (Total Adjustment Amount)	Estimated Giveback (Total Adjustment Amount - Pro-Rata)	# of Open Disputes
☐ West and Southwest	\$712,900	\$711,025	\$63,381	\$63,353	53
☐	\$380,558	\$380,558	\$292	\$292	16
☐	\$331,401	\$329,553	\$62,228	\$62,228	25
☐	\$816	\$816	\$816	\$816	4
☐	\$126	\$98	\$46	\$18	8
☐ Northeast and MidAtlantic	\$300,979	\$279,271	\$32,769	\$32,733	61
☐	\$108,062	\$87,327	\$61	\$24	32
☐	\$106,890	\$106,890	\$23,836	\$23,836	17
☐	\$75,353	\$74,380	\$8,873	\$8,873	9
☐	\$10,673	\$10,673	\$0	\$0	3
☐ Southeast and Midwest	\$282,350	\$282,350	\$14,757	\$14,757	33
☐	\$229,181	\$229,181	\$13,843	\$13,843	15
☐	\$53,170	\$53,170	\$914	\$914	12
☐	\$0	\$0	\$0	\$0	2
☐	\$0	\$0	\$0	\$0	4
Total	\$1,296,230	\$1,272,647	\$110,907	\$110,844	147

Open Disputes Summary: Divisional view with KPI cards showing count, avg days, and amounts by threshold

Sr. Leadership Dashboard: Tenant View

Same KPI card layout with the matrix switched to a Tenant-first hierarchy showing which tenants have the highest disputed amounts and most open cases across the portfolio.



Open Disputes Summary: Tenant view showing disputed amounts and case counts by tenant

Sr. Leadership Dashboard: Page Navigation

Hamburger menu navigation allowing users to move between the Open Disputes summary page and the Opened Disputes Trends section which keeps the interface clean and accessible for non-technical leadership users.

Tenant Disputes - Open Disputes

Tips for this Dashboard:

How do I quickly filter on the disputes related to my Division, Portfolio, etc.?

- To filter the card values to the right by Division, Regional Officer, or other descriptor, select the value in the matrix table below. For example, if I wanted the card values to adjust for the Northeast and MidAtlantic I would click "Northeast and MidAtlantic" in the table below. The selection would then highlight gray and update the card values.

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How do I view Open Dispute Trends?

- By clicking on the three bars or "hamburger" at the top right of the page, you will see additional pages available for you including an Opened Disputes graph trended over time.

Regional Officer View

Tenant View

of Disputes

of Disputes Open > \$20k

27

of Disputes

Avg Days Open Disputes > \$20k

132

Avg Days Open

Avg Amount Disputed > \$20k

\$38.16K

Avg Amount

Additional Pages

Open Disputes

Opened Disputes Trends

Tenant Name > Division > Shopping Center > Tenant # > Description > Issue Type > Year > Comments	Total Amount Disputed	Total Amount Disputed (Pro-Rata)	Estimated Giveback (Total Adjustment Amount)
101	\$380,267	\$380,267	
102	\$212,887	\$212,887	\$62,2
103	\$207,104	\$207,104	
104	\$105,800	\$105,800	
105	\$73,652	\$73,652	
106	\$70,126	\$70,126	\$30,4
107	\$53,519	\$53,519	\$22,3
108	\$40,642	\$40,642	
109	\$31,661	\$31,661	
110	\$27,700	\$27,700	
111	\$17,930	\$3,586	
112	\$12,427	\$12,427	\$0
113	\$11,640	\$11,640	
114	\$10,590	\$4,236	\$0
115	\$10,450	\$10,450	\$816
116	\$9,145	\$9,145	\$1,333
117	\$8,873	\$8,873	\$8,873
118	\$7,252	\$7,252	
Total	\$1,296,230	\$1,272,647	\$110,907

Page navigation menu: hamburger icon linking between dashboard sections

Sr. Leadership Dashboard: Drillthrough - Disputes Open >180 Days

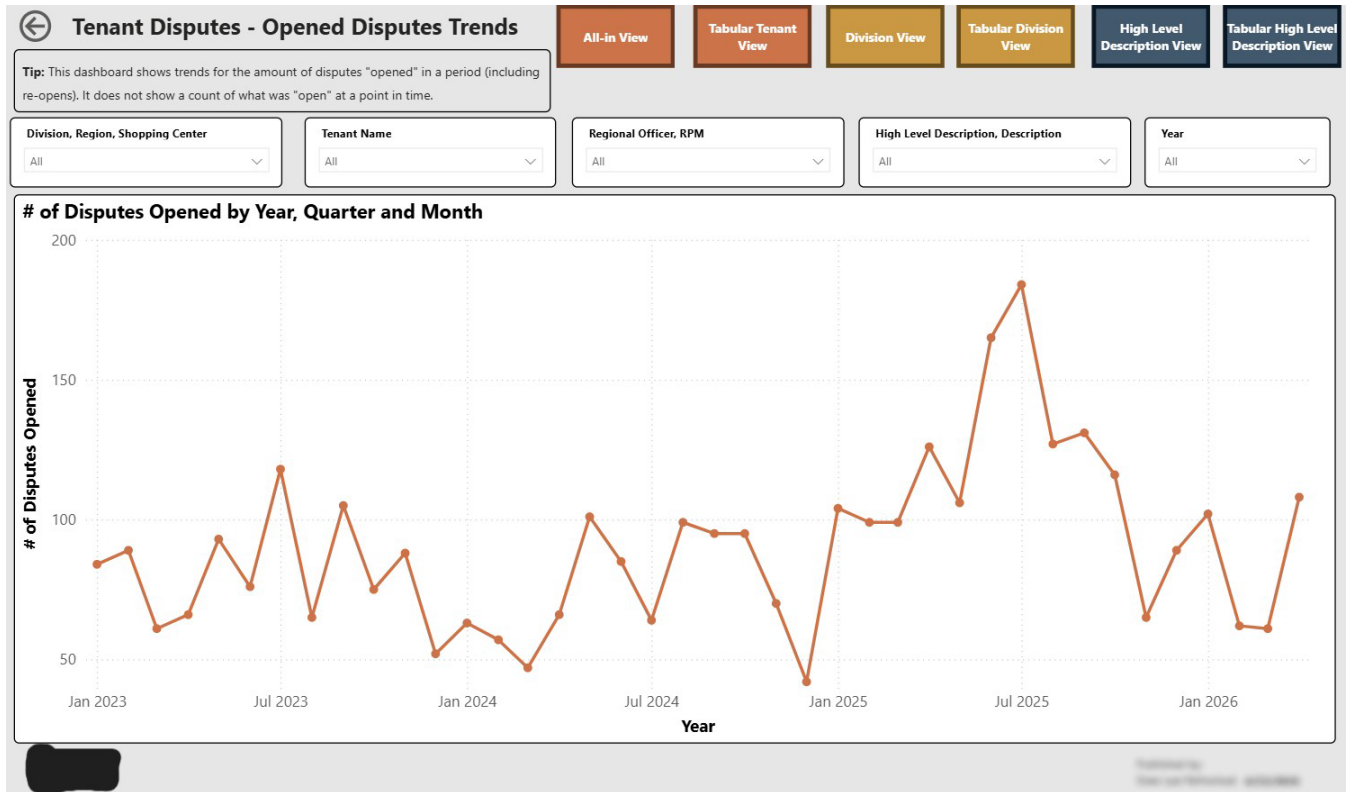
Drillthrough page accessed by right-clicking any KPI card value. Shows the full detail list of disputes meeting the selected threshold such as business unit, tenant, request ID, workflow status, description, issue type, billing type, days in status, and disputed amount.

Drillthrough - Disputes Open for Greater Than 180 Days																# of Disputes Open > 180 Days		
Division, Region, Shopping Center		Tenant Number		Tenant Name		Tenant Type		Description, Issue Type								Avg Days Open Disputes > 180 Days		
All		All		All		All		All								24		
EP PAM, EP PA		GL PAM, GL PA		Tenant Account Specialist		Regional Officer, RPM		Workflow Status								Avg Amount Disputed > 180 Days		
All		All		All		All		All								\$16.71K		
Business Unit	Shopping Center	Tenant Number	Tenant Name	Request ID (TDA)	Workflow Status	Description	Issue Type	Billing Type	Year	Today	Date Opened (TDA)	Start Date in Status	Days in Status	Days Open (TDA)	Total Amount Disputed	# of Disputes Open > 180 Days	Avg Days Open Disputes > 180 Days	Avg Amount Disputed (Open) Disputes > 180 Days
				4477	5 - With Legal	CAM PRIOR YR.	Audit	CAMP	2024	4/23/2026	9/22/2025	10/9/2025	196	213	\$66,756	1	213	\$66,756
				3862-1	1.1 - TDA Work In Progress	ANNUAL PAYER RET PRIOR YR	Audit,Lack of Documentation	AREP	2024	4/23/2026	8/11/2025	1/20/2026	93	255	\$41,767	1	255	\$41,767
				4401	4 - With TAS	REAL ESTATE TAX PRIOR YR.	Audit	RETP	2024	4/23/2026	9/1/2025	9/30/2025	205	234	\$26,868	1	234	\$26,868
				4400	4 - With TAS	REAL ESTATE TAX PRIOR YR.	Audit	RETP	2023	4/23/2026	9/1/2025	9/30/2025	205	234	\$26,614	1	234	\$26,614
				4399	4 - With TAS	REAL ESTATE TAX PRIOR YR.	Audit	RETP	2022	4/23/2026	9/1/2025	9/30/2025	205	234	\$26,247	1	234	\$26,247
				4398	4 - With TAS	REAL ESTATE TAX PRIOR YR.	Audit	RETP	2021	4/23/2026	9/1/2025	9/30/2025	205	234	\$26,072	1	234	\$26,072
				4472	5 - With Legal	CAM PRIOR YR.	Audit	CAMP	2019	4/23/2026	9/22/2025	10/9/2025	196	213	\$25,730	1	213	\$25,730
				4476	5 - With Legal	CAM PRIOR YR.	Audit	CAMP	2023	4/23/2026	9/22/2025	10/9/2025	196	213	\$25,424	1	213	\$25,424
				4475	5 - With Legal	CAM PRIOR YR.	Audit	CAMP	2022	4/23/2026	9/22/2025	10/9/2025	196	213	\$25,418	1	213	\$25,418
				4474	5 - With Legal	CAM PRIOR YR.	Audit	CAMP	2021	4/23/2026	9/22/2025	10/9/2025	196	213	\$24,025	1	213	\$24,025
				4327	1.1 - TDA Work In	ANNUAL PAYER RET PRIOR YR	Audit	AREP	2021	4/23/2026	8/11/2025	1/20/2026	93	255	\$23,240	1	255	\$23,240
Total															\$406,994	24	242	\$16,708

Drillthrough: Full detail for disputes open >180 days including timing and amounts

Trends: All-In Trended Graph View

Monthly count of disputes opened from January 2023 through May 2026. Allows leadership to spot seasonal patterns, identify months with unusual dispute volume, and measure the effectiveness of process improvements over time.



Opened Disputes Trends: Monthly count Jan 2023–May 2026 showing volume patterns over time

Trends: Tabular Tenant View

Tabular version of trends data broken out by tenant with quarterly totals from 2023 through 2026. Quickly identifies tenants with persistently high dispute volumes which is a leading indicator of lease or billing relationship issues.

Tenant Disputes - Opened Disputes Trends

All-in View

Tabular Tenant View

Division View

Tabular Division View

High Level Description View

Tabular High Level Description View

Tip: This dashboard shows trends for the amount of disputes "opened" in a period (including re-opens). It does not show a count of what was "open" at a point in time.

Division, Region, Shopping Center

Tenant Name

Regional Officer, RPM

High Level Description, Description

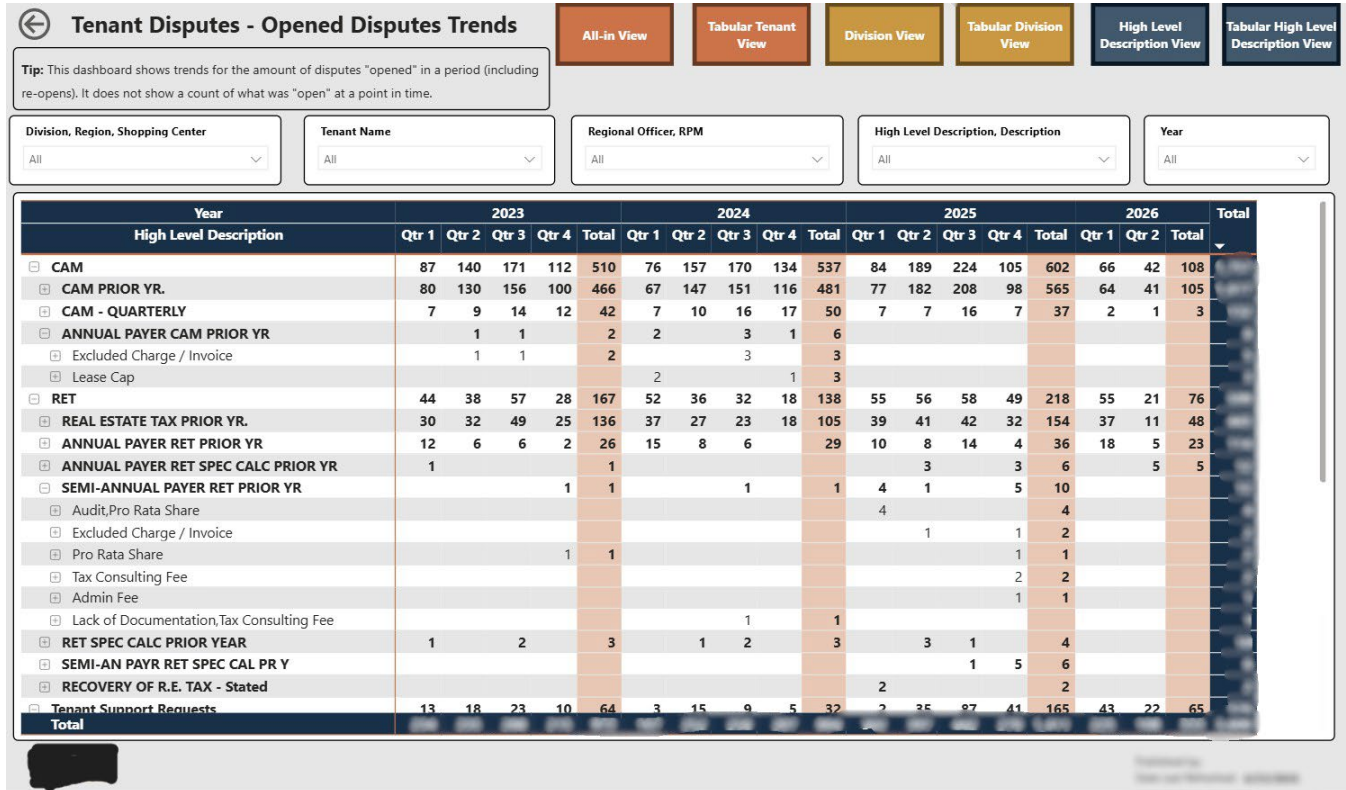
Year

Year	2023					2024					2025					2026			Total
TenantName	Qtr 1	Qtr 2	Qtr 3	Qtr 4	Total	Qtr 1	Qtr 2	Qtr 3	Qtr 4	Total	Qtr 1	Qtr 2	Qtr 3	Qtr 4	Total	Qtr 1	Qtr 2	Total	
CAM	40	11	14	27	92	6	11	9	10	36	21	13	22	16	72	12	6	18	
ESCROW or BUDGET Items	9	7	12	20	48	4	11	7	8	30	4	5	16	5	30	5	4	9	
RET	22			3	25				2	2	10	5			15	2	1	3	
RET - SUPP	8	3	2	2	15	2		2		4	6		4	3	13	3		3	
INS	1			2	3						1	2	1	4	8	2		2	
Tenant Support Requests		1			1								1	1	3				
PERCENTAGE RENT															2				
WASTE															1				
CAM	13	6	15	11	45	7	9	10	12	38	6	11	21	13	51	5	5	10	
RET	2	4	13	9	28	6	9	10	11	36	1	7	10	1	19	1	1	2	
ESCROW or BUDGET Items	3				3						3	2	9	5	19	1	4	5	
Tenant Support Requests	3	2	1	1	7	1			1	2	2			7	9	3		3	
RET - SUPP			1	1	2							2	2		4				
CAM	22	9	7	4	42	20	15	7	3	45	4	2		10	16	6	21	27	
INS	14	7	5	2	28	10	8	5	1	24		1			1	4	8	12	
RET	5		2	2	9	2	6	1	1	10			5		5	1	5	6	
Tenant Support Requests	1	2			3	7	1			8						8		8	
ESCROW or BUDGET Items								1		1		1		5	6	1		1	
Total											4				4				

Opened Disputes Trends: Tabular tenant view with quarterly breakdown 2023–2026

Trends: High-Level Description View

Tabular trends broken out by dispute description category (CAM, Real Estate Tax, Insurance, etc.) and sub-type. Enables management to identify which dispute categories are growing or shrinking over time.



Opened Disputes Trends: Category trends by quarter 2023–2026

Trends: Drillthrough Example

Drillthrough from any trend data point to the full underlying dispute list — including all relevant fields for each dispute matching the selection criteria.

⬅️ Drillthrough - Disputes Opened (All)

of Disputes Opened in 2025

of Disputes Opened in 2026

Division, Region, Shopping Center

All

Tenant Number

All

Tenant Name

All

Tenant Type

All

Description, Issue Type

All

EP PAM, EP PA

All

GL PAM, GL PA

All

Tenant Account Specialist

All

Regional Officer, RPM

All

Workflow Status

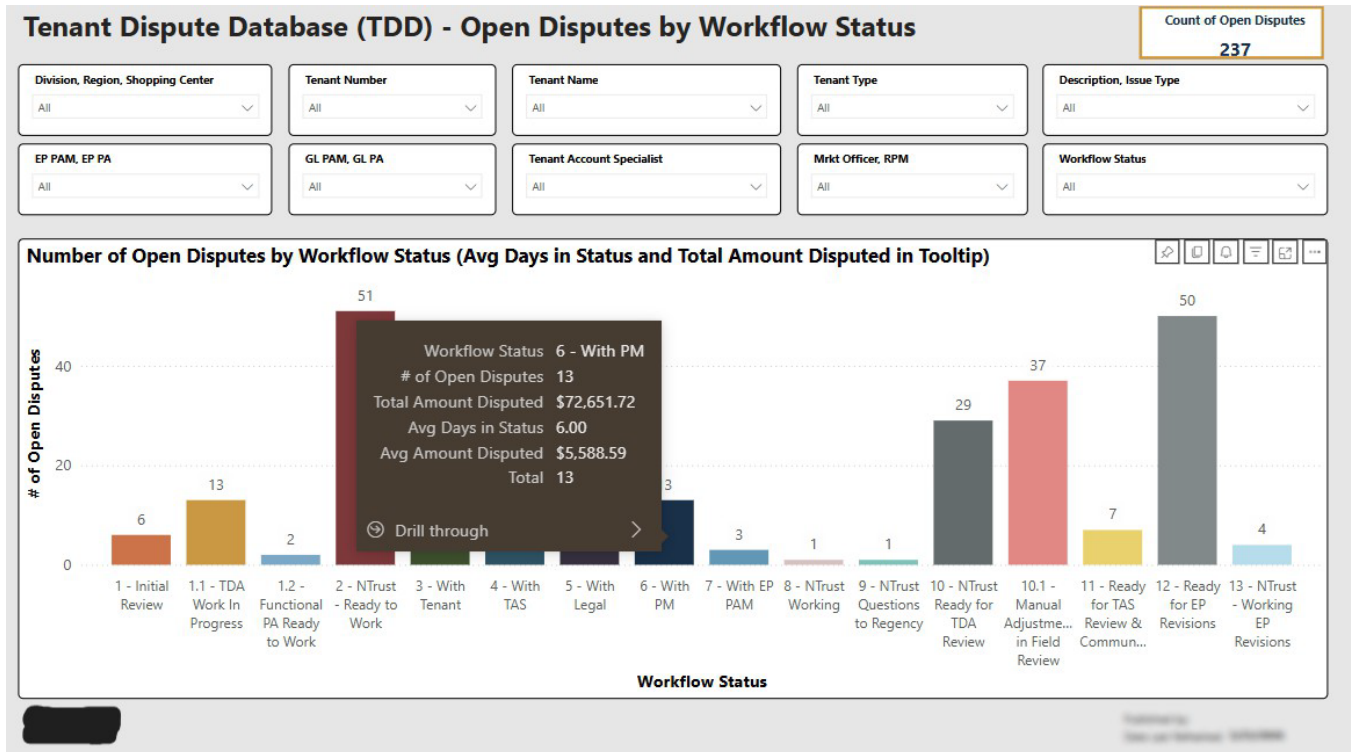
All

BusinessUnit	ShoppingCenter	TenantNumber	TenantName	Request ID (TDA)	Date Opened (TDA)	Description	Issue Type	Billing Type	Workflow Status	Total Amount Disputed
1000	Westfield Shopping Center	10000	10000000	1155-1	1/11/2023	CAM PRIOR YR.	Excluded Charge / Invoice, Lease Cap	CAMP	Closed	\$21,721
1000	Westfield Shopping Center	10000	10000000	1165-1	1/4/2023	CAM PRIOR YR.	Audit	CAMP	Closed	\$1,965
1000	Westfield Shopping Center	10000	10000000	1165-2	4/12/2023	CAM PRIOR YR.	Audit	CAMP	Closed	\$1,965
1000	Westfield Shopping Center	10000	10000000	1166-1	1/4/2023	CAM PRIOR YR.	Audit	CAMP	Closed	\$3,036
1000	Westfield Shopping Center	10000	10000000	1166-2	3/15/2023	CAM PRIOR YR.	Audit	CAMP	Closed	\$3,036
1000	Westfield Shopping Center	10000	10000000	1166-3	4/12/2023	CAM PRIOR YR.	Audit	CAMP	Closed	\$3,036
1000	Westfield Shopping Center	10000	10000000	1198-1	1/8/2023	CAM PRIOR YR.	Audit, Lease Cap	CAMP	Closed	\$6,225
1000	Westfield Shopping Center	10000	10000000	1198-2	5/30/2023	CAM PRIOR YR.	Audit, Lease Cap	CAMP	Closed	\$6,225
1000	Westfield Shopping Center	10000	10000000	1295-1	4/30/2023	CAM PRIOR YR.	Excluded Charge / Invoice	CAMP	Closed	\$1,247
1000	Westfield Shopping Center	10000	10000000	1370-1	11/26/2024	CAM PRIOR YR.	Excluded Charge / Invoice, Management Fee	CAMP	Closed	\$698
1000	Westfield Shopping Center	10000	10000000	1380-1	5/9/2023	CAM PRIOR YR.	Lease Cap	CAMP	Closed	\$1,765
1000	Westfield Shopping Center	10000	10000000	1380-2	6/5/2023	CAM PRIOR YR.	Lease Cap	CAMP	Closed	\$1,765
1000	Westfield Shopping Center	10000	10000000	1408-1	5/6/2024	CAM PRIOR YR.	Excluded Charge / Invoice, Lease Cap	CAMP	Closed	\$1,532
1000	Westfield Shopping Center	10000	10000000	1433-1	10/26/2023	CAM PRIOR YR.	Excluded Charge / Invoice	CAMP	Closed	(\$15,944)
1000	Westfield Shopping Center	10000	10000000	1459-1	6/26/2024	CAM PRIOR YR.	Admin Fee	CAMP	Closed	(\$345)
1000	Westfield Shopping Center	10000	10000000	1459-2	6/28/2024	CAM PRIOR YR.	Admin Fee	CAMP	Closed	(\$345)
1000	Westfield Shopping Center	10000	10000000	1463-1	12/19/2023	CAM PRIOR YR.	Lease Cap	CAMP	Closed	\$375
1000	Westfield Shopping Center	10000	10000000	1533-1	9/6/2023	CAM PRIOR YR.	Excluded Charge / Invoice	CAMP	Closed	\$2,983
Total										\$21,721

Trends drillthrough: Full dispute detail list accessible from any data point

Operational Dashboard: Open Disputes Graph View

Bar chart of open disputes by workflow status. Tooltip on hover shows dispute count, total amount, average days in status, and average amount per dispute for that workflow stage. Drill-through available from each bar.



Operational view: Open disputes by workflow status with rich tooltip detail and drillthrough

Operational Dashboard: Table View & Drillthrough

Table view showing all open disputes sorted by days in status (oldest first). Accessed directly or via drillthrough from the bar chart. Shows full dispute detail including workflow status, description, issue type, billing type, and days open.

← Tenant Dispute Database (TDD) - Open Disputes Status

Count of Open Disputes

237

Division, Region, Shopping Center

All

Tenant Number

All

Tenant Name

All

Tenant Type

All

Description, Issue Type

All

EP PAM, EP PA

All

GL PAM, GL PA

All

Tenant Account Specialist

All

Mkt Officer, RPM

All

Workflow Status

All

Business Unit	Shopping Center	Tenant Number	TenantName	Request ID	Assigned To	Workflow Status	Description	Issue Type	BillingType	Year	Start Date in Status	Today	Days in Status	Total Amount Disputed
				3717		5 - With Legal	CAM PRIOR YR.	Excluded Charge / Invoice,Lack of Documentation,Pro Rata Share	CAMP	2023	4/6/2025	5/23/2026	412	\$0.00
				3715		13 - NTrust - Working EP Revisions	CAM PRIOR YR.	Excluded Charge / Invoice,Lack of Documentation,Lease Cap,Pro Rata Share	CAMP	2023	1/5/2026	5/23/2026	138	\$4,167.79
				3724		13 - NTrust - Working EP Revisions	INSURANCE PRIOR YEAR	Excluded Charge / Invoice,Lack of Documentation	INSP	2023	1/5/2026	5/23/2026	138	\$5,979.00
				4820		1.2 - Functional PA Ready to Work	ESCROW or BUDGET Items	Excluded Charge / Invoice	ESCROWS	2026	1/25/2026	5/23/2026	118	\$0.00
				4879		2 - NTrust - Ready to Work	REAL ESTATE TAX PRIOR YR.	Lack of Documentation	RETP	2025	2/22/2026	5/23/2026	90	\$3,146.76
				3906		4 - With TAS	PERCENTAGE RENT	Excluded Charge / Invoice	PCTR	2023	2/23/2026	5/23/2026	89	\$1,920.71
				4020		4 - With TAS	PERCENTAGE RENT	Excluded Charge / Invoice	PCTR	2022	2/23/2026	5/23/2026	89	\$2,007.89
				4522		4 - With TAS	PERCENTAGE RENT	Excluded Charge / Invoice	PCTR	2023	2/23/2026	5/23/2026	89	\$4,020.60
				4523		4 - With TAS	PERCENTAGE RENT	Excluded Charge / Invoice	PCTR	2024	2/23/2026	5/23/2026	89	\$4,478.02
				4330		4 - With TAS	INSURANCE PRIOR YEAR	Lease Cap	INSP	2024	3/2/2026	5/23/2026	82	(\$5,199.09)
				4855		3 - With EP PAM	ESCROW - BUDGET	Lack of Documentation	ESCROWS	2025	3/3/2026	5/23/2026	82	\$0.00
Total														\$1,583,395.54

Open Disputes Table: Full detail view sortable by any column, with drillthrough from bar chart

Operational Dashboard: Drillthrough to Ticket Comments

Additional drillthrough layer accessing the comments and feedback notes on each dispute ticket. Allows the Disputes Manager to review communication history and outstanding questions directly in Power BI without switching to the source system.



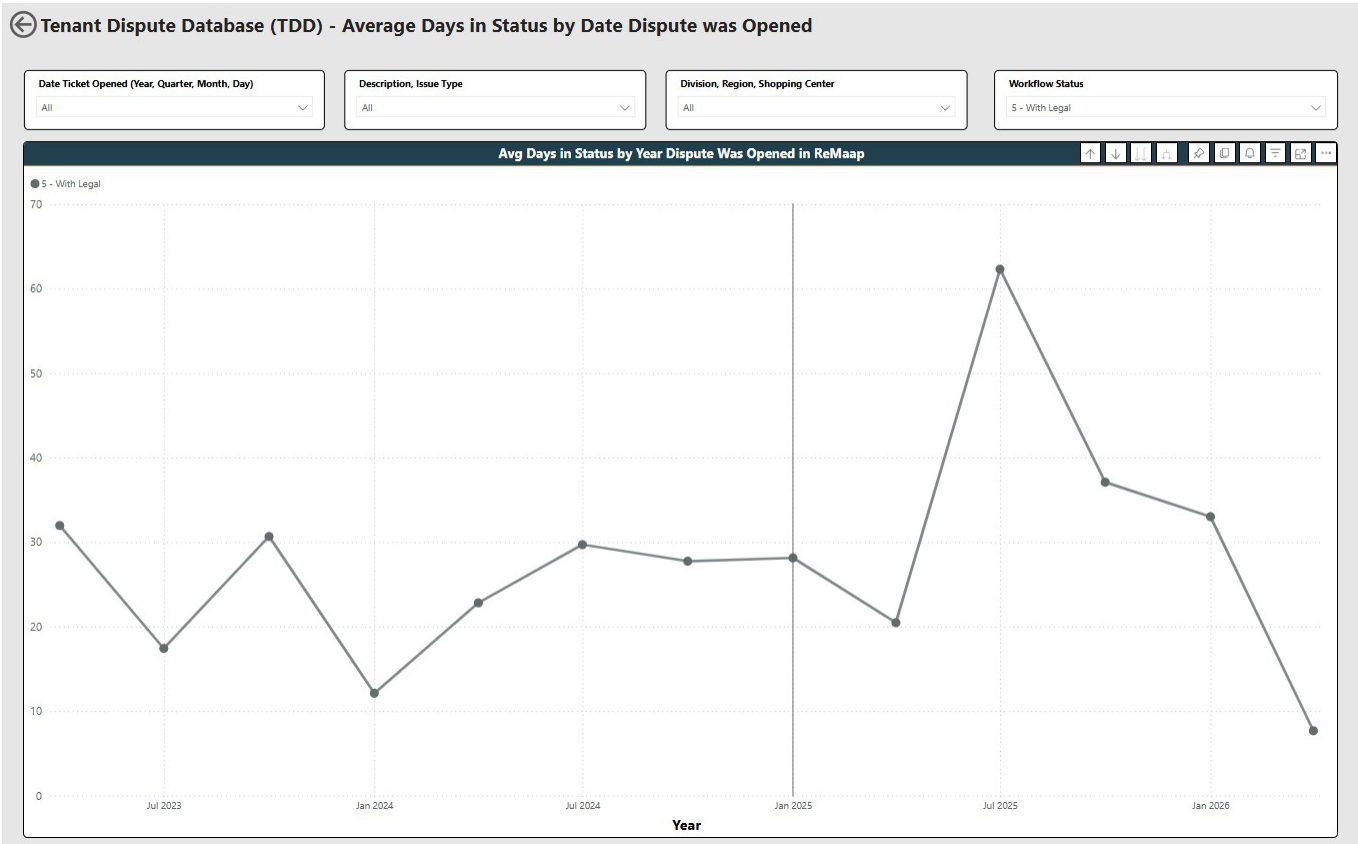
Drillthrough - Comments

ShoppingCenter	TenantNumber	TenantName	RequestID	Description	IssueType	Workflow Status	ModifiedDate	Comments	FeedbackToNTrust
Shopping Center 1	123456	Tenant ABC	3717	CAM PRIOR YR.	Excluded Charge / Invoice,Lack of Documentation,Pro Rata Share	5 - With Legal	2023-10-27 10:00:00	My concern is, have we been billing the outparcel occupant(s) per the Declaration?	

Drillthrough to Comments: Ticket-level comments and feedback visible within the dashboard

Operational Dashboard: Average Days in Status

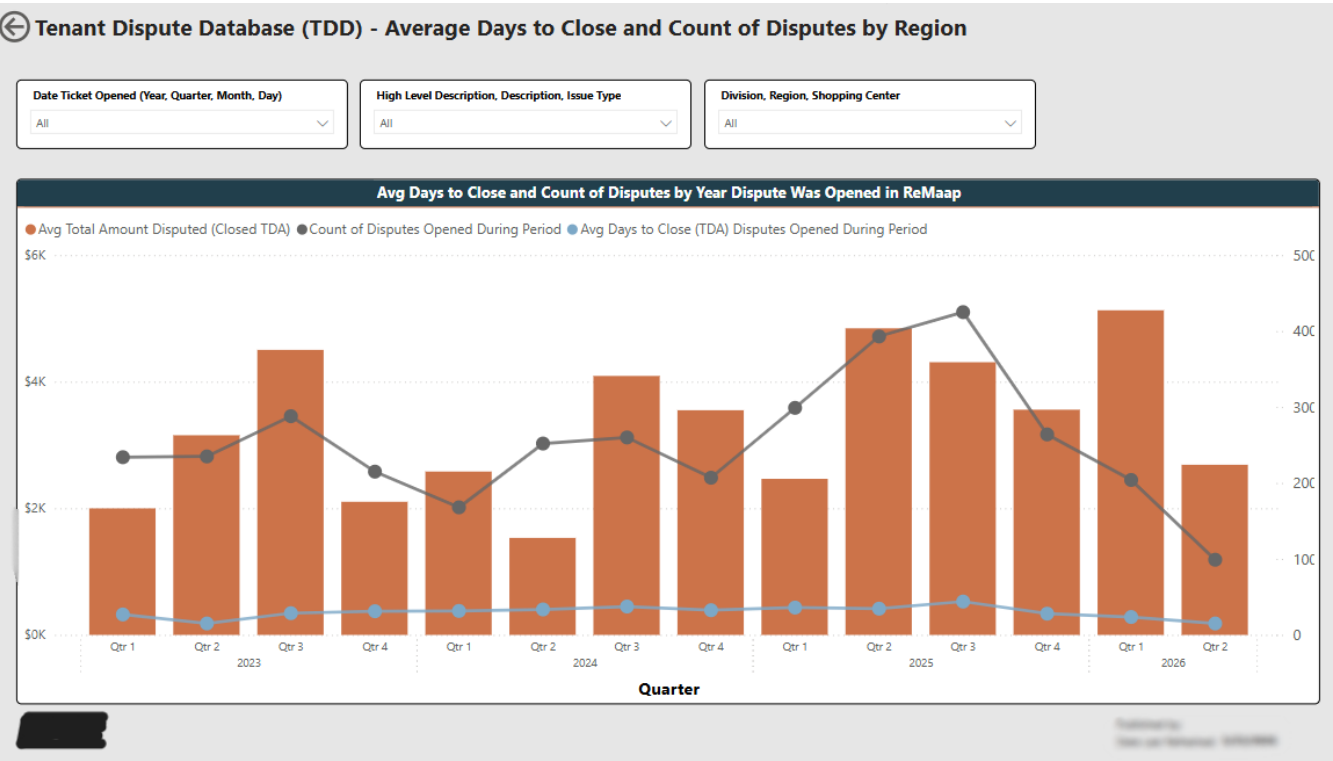
Trend chart showing average days disputes spend in a given workflow status. Helps identify bottlenecks (i.e. which stages are taking longer than expected and in which time periods).



Average Days in Status:Trend analysis identifying workflow bottlenecks by stage and period

Operational Dashboard: Average Days to Close

Combo chart showing average days to close alongside dispute count and average amount — trended by quarter from 2023 through 2026. Measures whether process improvements are reducing close times over time.



Average Days to Close: Quarterly trend of close speed, count, and average amount 2023–2026

Operational Dashboard: Closed Disputes Table

Full detail table of closed disputes showing request ID, workflow status, description, issue type, dates opened and closed, days to close, total adjustment amount, and total amount. Filterable by all standard dimensions. Amounts shown are a filtered subset.

←

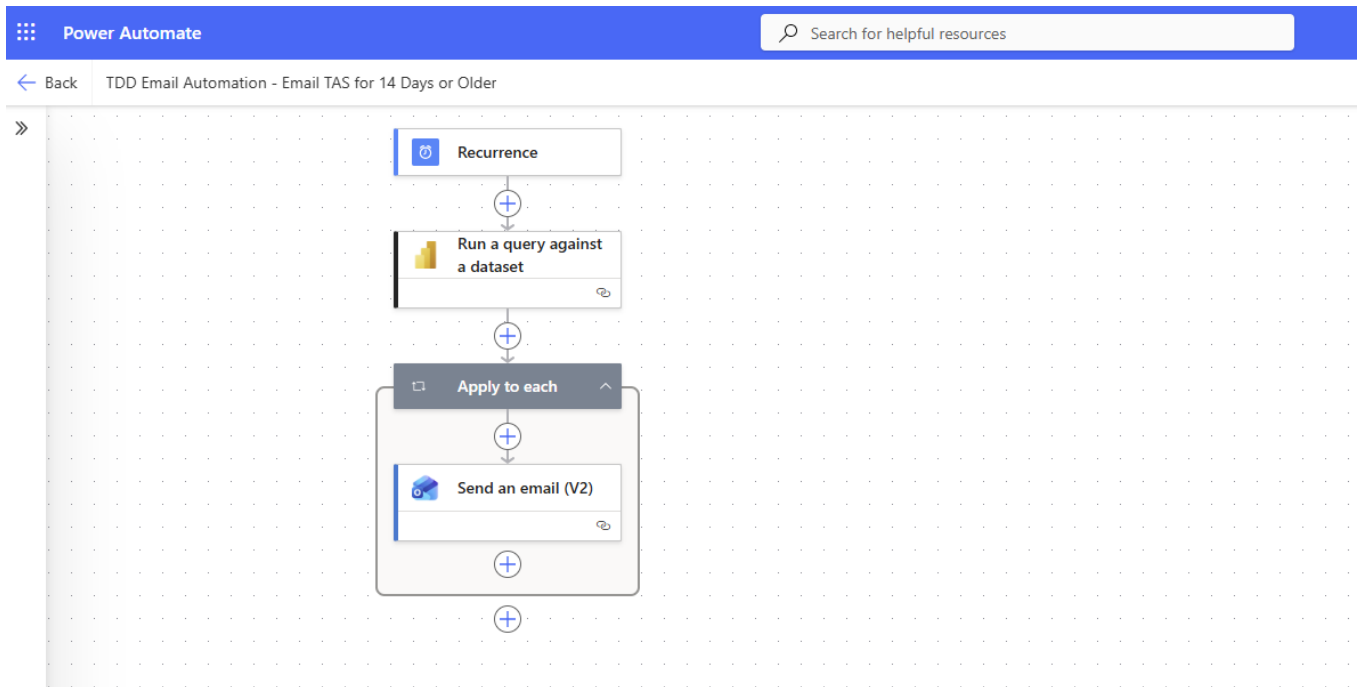
Tenant Dispute Database (TDD) - Closed Tickets

Business Unit	Shopping Center	Tenant Number	Tenant Name	RequestID (Modified)	Description	IssueType	Status	Date Opened (TDA)	Date Closed (TDA)	Days To Close (TDA)	Total Adjustment Amount	Total Amount Disputed
100	Shopping Center	1000	Tenant A	1155-1	CAM PRIOR YR.	Excluded Charge / Invoice, Lease Cap	Closed	1/11/2023	2/14/2023	34	0.00	21,720.82
100	Shopping Center	1000	Tenant A	1165-1	CAM PRIOR YR.	Audit	Closed	1/4/2023	2/14/2023	41	0.00	1,964.72
100	Shopping Center	1000	Tenant A	1165-2	CAM PRIOR YR.	Audit	Closed	4/12/2023	4/19/2023	6	0.00	1,964.72
100	Shopping Center	1000	Tenant A	1166-1	CAM PRIOR YR.	Audit	Closed	1/4/2023	2/14/2023	41	0.00	3,035.71
100	Shopping Center	1000	Tenant A	1166-2	CAM PRIOR YR.	Audit	Closed	3/15/2023	3/16/2023	0	0.00	3,035.71
100	Shopping Center	1000	Tenant A	1166-3	CAM PRIOR YR.	Audit	Closed	4/12/2023	4/19/2023	6	0.00	3,035.71
100	Shopping Center	1000	Tenant A	1198-1	CAM PRIOR YR.	Audit, Lease Cap	Closed	1/8/2023	2/20/2023	43	0.00	6,225.34
100	Shopping Center	1000	Tenant A	1198-2	CAM PRIOR YR.	Audit, Lease Cap	Closed	5/30/2023	5/31/2023	0	0.00	6,225.34
100	Shopping Center	1000	Tenant A	1241-1	INSURANCE PRIOR YEAR	Pro Rata Share	Closed	5/10/2023	5/10/2023	0	0.00	2,198.78
100	Shopping Center	1000	Tenant A	1247-1	ANNUAL PAYER RET PRIOR YR.	Audit	Closed	1/21/2026	2/3/2026	12	16,343.98	16,344.00
100	Shopping Center	1000	Tenant A	1275-1	REAL ESTATE TAX PRIOR YR.	Tax Consulting Fee	Closed	11/20/2023	11/20/2023	0	0.00	57.05
100	Shopping Center	1000	Tenant A	1295-1	CAM PRIOR YR.	Excluded Charge / Invoice	Closed	4/30/2023	5/11/2023	10	0.00	1,246.67
100	Shopping Center	1000	Tenant A	1370-1	CAM PRIOR YR.	Excluded Charge / Invoice, Management Fee	Closed	11/26/2024	12/15/2024	18	697.79	697.79
100	Shopping Center	1000	Tenant A	1380-1	CAM PRIOR YR.	Lease Cap	Closed	5/9/2023	5/30/2023	21	0.00	1,765.35
100	Shopping Center	1000	Tenant A	1380-2	CAM PRIOR YR.	Lease Cap	Closed	6/5/2023	6/19/2023	13	0.00	1,765.35
100	Shopping Center	1000	Tenant A	1408-1	CAM PRIOR YR.	Excluded Charge / Invoice, Lease Cap	Closed	5/6/2024	5/16/2024	9	0.00	1,531.79
100	Shopping Center	1000	Tenant A	1433-1	CAM PRIOR YR.	Excluded Charge / Invoice	Closed	10/26/2023	11/19/2023	24	0.00	-15,943.58
100	Shopping Center	1000	Tenant A	1448-1	REAL ESTATE TAX PRIOR YR.	Pro Rata Share	Closed	3/26/2023	6/25/2023	91	0.00	234.35
100	Shopping Center	1000	Tenant A	1459-1	CAM PRIOR YR.	Admin Fee	Closed	6/26/2024	6/26/2024	0	-345.26	-345.26
100	Shopping Center	1000	Tenant A	1459-2	CAM PRIOR YR.	Admin Fee	Closed	6/28/2024	7/29/2024	31	-345.26	-345.26
100	Shopping Center	1000	Tenant A	1463-1	CAM PRIOR YR.	Lease Cap	Closed	12/19/2023	1/18/2024	29	0.00	375.17
100	Shopping Center	1000	Tenant A	1464-1	Tenant Support Requests	Lack of Documentation	Closed	1/5/2023	2/15/2023	40	0.00	0.00
100	Shopping Center	1000	Tenant A	1474-1	INSURANCE PRIOR YEAR	Excluded Charge /	Closed	10/18/2023	10/18/2023	0	0.00	1,537.78

Closed Disputes Table: Full history with close timing, adjustment amounts, and resolution details

Email Automation: Power Automate Cloud Flow

Scheduled Power Automate Cloud flow that runs automatically. Queries the Power BI semantic model via the Execute Dataset Query action to identify disputes meeting each escalation threshold, then loops through results and sends personalized reminder emails to each responsible party. Several flows were created for 48 hour reminders, 14 day reminders, and 30+ day reminders with direct supervisor CC'd.



Email Automation: DAX Query Code

DAX query embedded in the Power Automate flow. Filters disputes by workflow status and days in status (>13 days), then selects the required fields including assignee email for the automated send.

 **Power Automate**

 Back

TDD Email Automation - Email TAS for 14 Days or Older

 Run a query against a dataset

Parameters

Settings

Code view

Testing

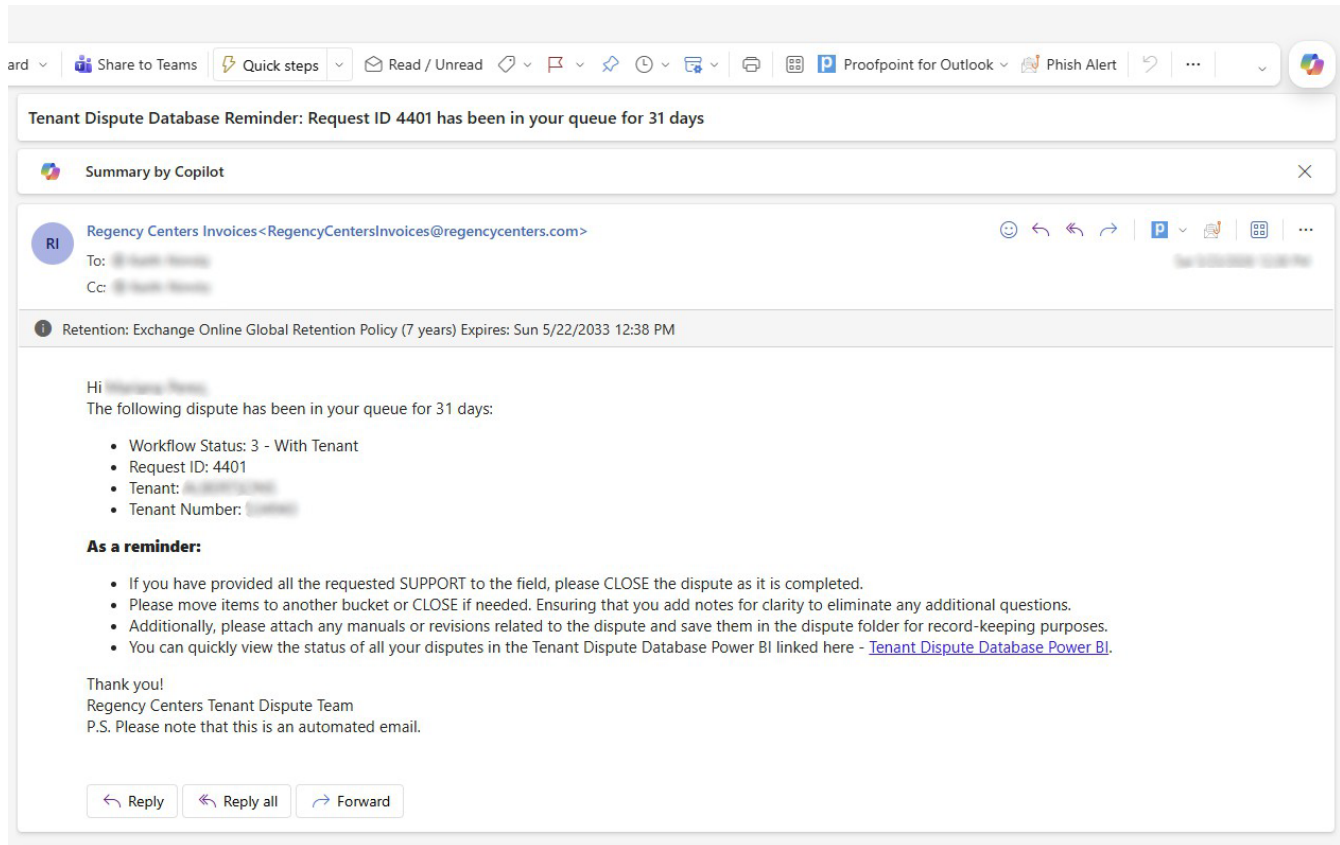
About

```
1 {
2   "type": "OpenApiConnection",
3   "inputs": {
4     "parameters": {
5       "groupid": "42dbb0b6-7949-4738-b2db-b05ee626626c",
6       "datasetid": "5852cfc6-116f-43d0-8437-10ff5016547b",
7       "specification/query": "EVALUATE SELECTCOLUMNS( FILTER( 'ADF
vTenantDispute', 'ADF vTenantDispute'[Days With (Duration)] >13 &&
'ADF vTenantDispute'[WorkflowStatus] <> \"Closed\" && 'ADF
vTenantDispute'[WorkflowStatus] IN { \"3 - With Tenant\", \"4 -
With TAS\", \"11 - Ready for TAS Review & Communication\" } ),
\"RequestID\", 'ADF vTenantDispute'[RequestID], \"AssignedTo\",
'ADF vTenantDispute'[AssignedTo], \"TenantName\", 'ADF
vTenantDispute'[TenantName], \"TenantNumber\", 'ADF vTenantDispute'
[TenantNumber], \"DaysInStatus\", 'ADF vTenantDispute'[Days With
(Duration)], \"WorkflowStatus\", 'ADF vTenantDispute'
[WorkflowStatus], \"TAS Specialist\", 'ADF vTenantDispute'
[TenantAccountSpecialist], \"TAS Email\", 'ADF vTenantDispute'[TAS
Email] )",
8       "specification/serializerSettings/includeNulls": false
9     },
10    "host": {
11      "apiId": "/providers/Microsoft.PowerApps/apis/shared_powerbi",
12      "connection": "shared_powerbi-1",
13      "operationId": "ExecuteDatasetQuery"
14    }
15  },
16  "runAfter": {}
17 }
```

DAX query: Filters active disputes by status and aging threshold, returns assignee contact fields

Email Automation: Example Automated Email

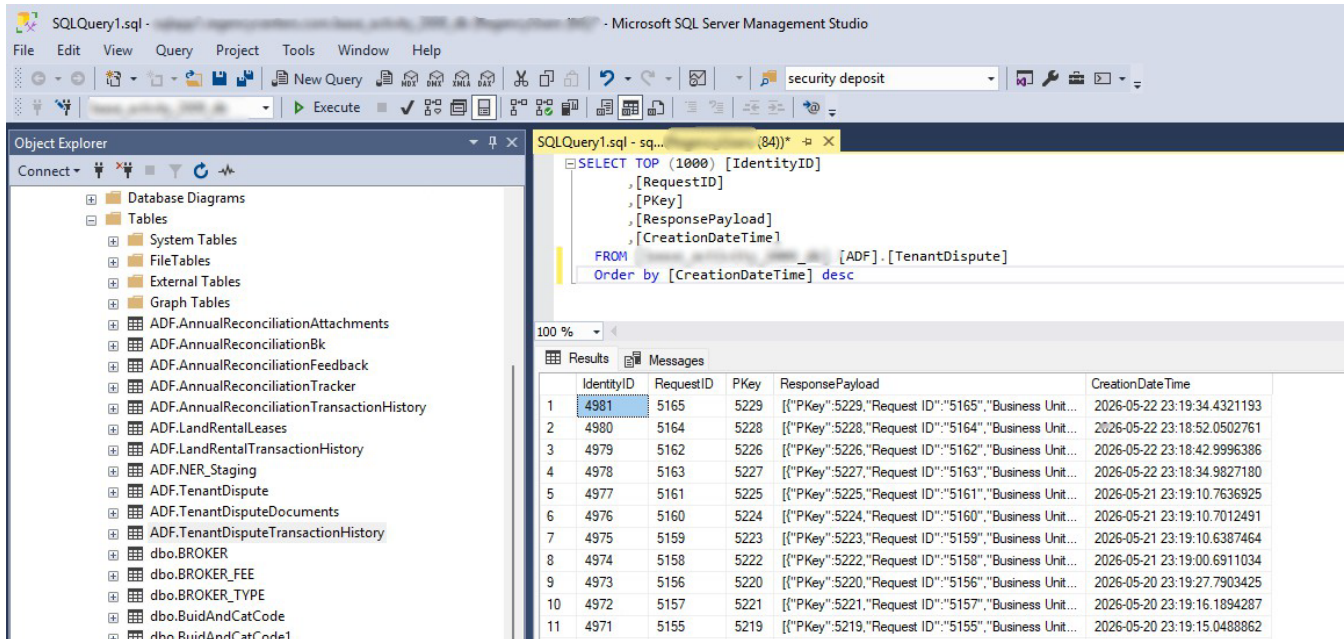
Sample of the automated reminder email sent to a Tenant Account Specialist when a dispute has been in their queue for 14+ days. Includes workflow status, request ID, tenant name, and action reminders are sent automatically without any manual intervention.



Example automated reminder email sent to responsible party with dispute details and action guidance

SQL Validation: Transaction History Table

SQL query against ADF.TenantDisputeTransactionHistory. Data validation and coordination in setting up the table herein, as well as, the other tables noted in this project was completed throughout execution utilizing SQL Server Management Studio to verify data loaded properly and continued to do so. A few changes were made from such coordination to ensure the data load was properly set-up.



The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure, including tables like ADF.TenantDisputeTransactionHistory. The central query window shows a SQL query selecting the top 1000 records from the ADF.TenantDisputeTransactionHistory table, ordered by CreationDateTime in descending order. The Results pane at the bottom shows the output of the query, displaying columns: IdentityID, RequestID, PKey, ResponsePayload, and CreationDateTime. The first row is highlighted.

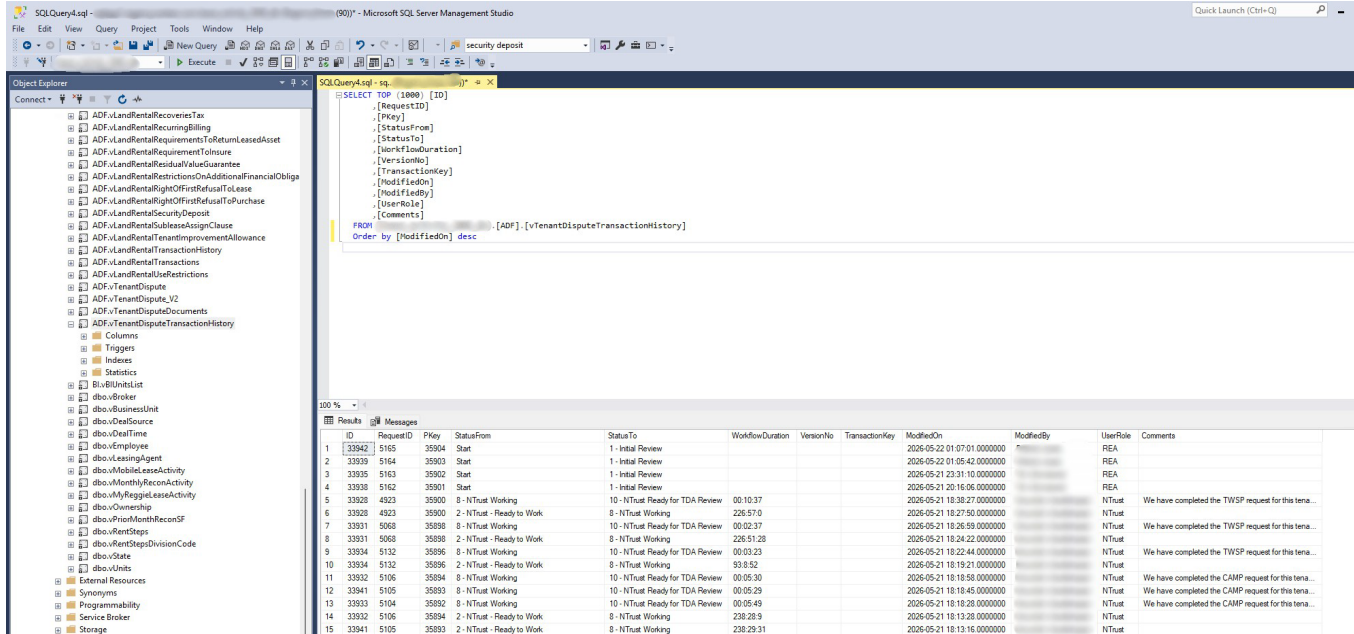
```
SELECT TOP (1000) [IdentityID]
, [RequestID]
, [PKey]
, [ResponsePayload]
, [CreationDateTime]
FROM [ADF].[TenantDispute]
Order by [CreationDateTime] desc
```

	IdentityID	RequestID	PKey	ResponsePayload	CreationDateTime
1	4981	5165	5229	[{"PKey":5229,"Request ID":"5165","Business Unit..."}	2026-05-22 23:19:34.4321193
2	4980	5164	5228	[{"PKey":5228,"Request ID":"5164","Business Unit..."}	2026-05-22 23:18:52.0502761
3	4979	5162	5226	[{"PKey":5226,"Request ID":"5162","Business Unit..."}	2026-05-22 23:18:42.9996386
4	4978	5163	5227	[{"PKey":5227,"Request ID":"5163","Business Unit..."}	2026-05-22 23:18:34.9827180
5	4977	5161	5225	[{"PKey":5225,"Request ID":"5161","Business Unit..."}	2026-05-21 23:19:10.7636925
6	4976	5160	5224	[{"PKey":5224,"Request ID":"5160","Business Unit..."}	2026-05-21 23:19:10.7012491
7	4975	5159	5223	[{"PKey":5223,"Request ID":"5159","Business Unit..."}	2026-05-21 23:19:10.6387464
8	4974	5158	5222	[{"PKey":5222,"Request ID":"5158","Business Unit..."}	2026-05-21 23:19:00.6911034
9	4973	5156	5220	[{"PKey":5220,"Request ID":"5156","Business Unit..."}	2026-05-20 23:19:27.7903425
10	4972	5157	5221	[{"PKey":5221,"Request ID":"5157","Business Unit..."}	2026-05-20 23:19:16.1894287
11	4971	5155	5219	[{"PKey":5219,"Request ID":"5155","Business Unit..."}	2026-05-20 23:19:15.0488862

SQL: ADF.TenantDisputeTransactionHistory table structure and sample records

SQL Validation: Transaction History View Table

Query against ADF.vTenantDisputeTransactionHistory showing workflow transition records including StatusFrom, StatusTo, WorkflowDuration, and Comments, the data that drives the average days and escalation logic in the dashboards.



```
SELECT TOP (1000) [ID],
                [RequestID],
                [PKey],
                [StatusFrom],
                [StatusTo],
                [WorkflowDuration],
                [VersionNo],
                [TransactionKey],
                [ModifiedOn],
                [ModifiedBy],
                [UserRole],
                [Comments]
FROM [ADF].[vTenantDisputeTransactionHistory]
ORDER BY [ModifiedOn] desc
```

ID	RequestID	PKey	StatusFrom	StatusTo	WorkflowDuration	VersionNo	TransactionKey	ModifiedOn	ModifiedBy	UserRole	Comments
1	33942	5165	35904	Start	1 - Initial Review			2026-05-22 01:07:01.0000000		REA	
2	33939	5164	35903	Start	1 - Initial Review			2026-05-22 01:05:42.0000000		REA	
3	33935	5163	35902	Start	1 - Initial Review			2026-05-21 23:31:10.0000000		REA	
4	33938	5162	35901	Start	1 - Initial Review			2026-05-21 20:16:06.0000000		REA	
5	33928	4923	35900	8 - N-Trust - Ready to Work	10 - N-Trust Ready for TDA Review	00:10:37		2026-05-21 18:38:27.0000000		NTrust	We have completed the TWSP request for this tena...
6	33928	4923	35900	2 - N-Trust - Ready to Work	8 - N-Trust Working	226:57:0		2026-05-21 18:27:50.0000000		NTrust	
7	33931	5068	35898	8 - N-Trust Working	10 - N-Trust Ready for TDA Review	00:02:37		2026-05-21 18:26:59.0000000		NTrust	We have completed the TWSP request for this tena...
8	33931	5068	35898	2 - N-Trust - Ready to Work	8 - N-Trust Working	226:51:28		2026-05-21 18:24:22.0000000		NTrust	
9	33934	5132	35896	8 - N-Trust Working	10 - N-Trust Ready for TDA Review	00:03:23		2026-05-21 18:22:44.0000000		NTrust	We have completed the TWSP request for this tena...
10	33934	5132	35896	2 - N-Trust - Ready to Work	8 - N-Trust Working	93:8:52		2026-05-21 18:19:21.0000000		NTrust	
11	33932	5106	35894	8 - N-Trust Working	10 - N-Trust Ready for TDA Review	00:05:30		2026-05-21 18:18:58.0000000		NTrust	We have completed the CAMP request for this tena...
12	33941	5105	35893	8 - N-Trust Working	10 - N-Trust Ready for TDA Review	00:05:29		2026-05-21 18:18:45.0000000		NTrust	We have completed the CAMP request for this tena...
13	33933	5104	35892	8 - N-Trust Working	10 - N-Trust Ready for TDA Review	00:05:49		2026-05-21 18:18:28.0000000		NTrust	We have completed the CAMP request for this tena...
14	33932	5106	35894	2 - N-Trust - Ready to Work	8 - N-Trust Working	238:28:9		2026-05-21 18:13:28.0000000		NTrust	
15	33941	5105	35893	2 - N-Trust - Ready to Work	8 - N-Trust Working	238:29:31		2026-05-21 18:13:16.0000000		NTrust	

SQL: vTenantDisputeTransactionHistory view with status transitions, timing, and comments

SQL Validation: Dispute Table

Query against ADF.TenantDispute confirming dispute master records including the ResponsePayload JSON structure from the API. The table was examined and validated that the full dispute record is loading correctly and all fields needed for Power BI are present.

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows a list of databases, with 'ADF.TenantDispute' selected. The query window on the right contains the following SQL query:

```
SELECT TOP (1000) [IdentityID]
, [RequestID]
, [PKey]
, [ResponsePayload]
, [CreationDateTime]
FROM [ADF].[TenantDispute]
Order by [CreationDateTime] desc
```

The Results tab at the bottom shows the output of the query, displaying columns: IdentityID, RequestID, PKey, ResponsePayload, and CreationDateTime. The data is sorted by CreationDateTime in descending order.

	IdentityID	RequestID	PKey	ResponsePayload	CreationDateTime
1	4981	5165	5229	[{"PKey": "5229", "Request ID": "5165", "Business Unit": "5165"}]	2026-05-22 23:19:34.4321193
2	4980	5164	5228	[{"PKey": "5228", "Request ID": "5164", "Business Unit": "5164"}]	2026-05-22 23:18:52.0502761
3	4979	5162	5226	[{"PKey": "5226", "Request ID": "5162", "Business Unit": "5162"}]	2026-05-22 23:18:42.9996386
4	4978	5163	5227	[{"PKey": "5227", "Request ID": "5163", "Business Unit": "5163"}]	2026-05-22 23:18:34.9827180
5	4977	5161	5225	[{"PKey": "5225", "Request ID": "5161", "Business Unit": "5161"}]	2026-05-21 23:19:10.7636925
6	4976	5160	5224	[{"PKey": "5224", "Request ID": "5160", "Business Unit": "5160"}]	2026-05-21 23:19:10.7012491
7	4975	5159	5223	[{"PKey": "5223", "Request ID": "5159", "Business Unit": "5159"}]	2026-05-21 23:19:10.6387464

SQL: ADF.TenantDispute table showing RequestID, PKey, and ResponsePayload from API

SQL Validation: Dispute View Table

Query against ADF.vTenantDispute_V2. This view feeds the full cross-dimensional filtering capability in Power BI.

SQL Query2.sql - (1177)*

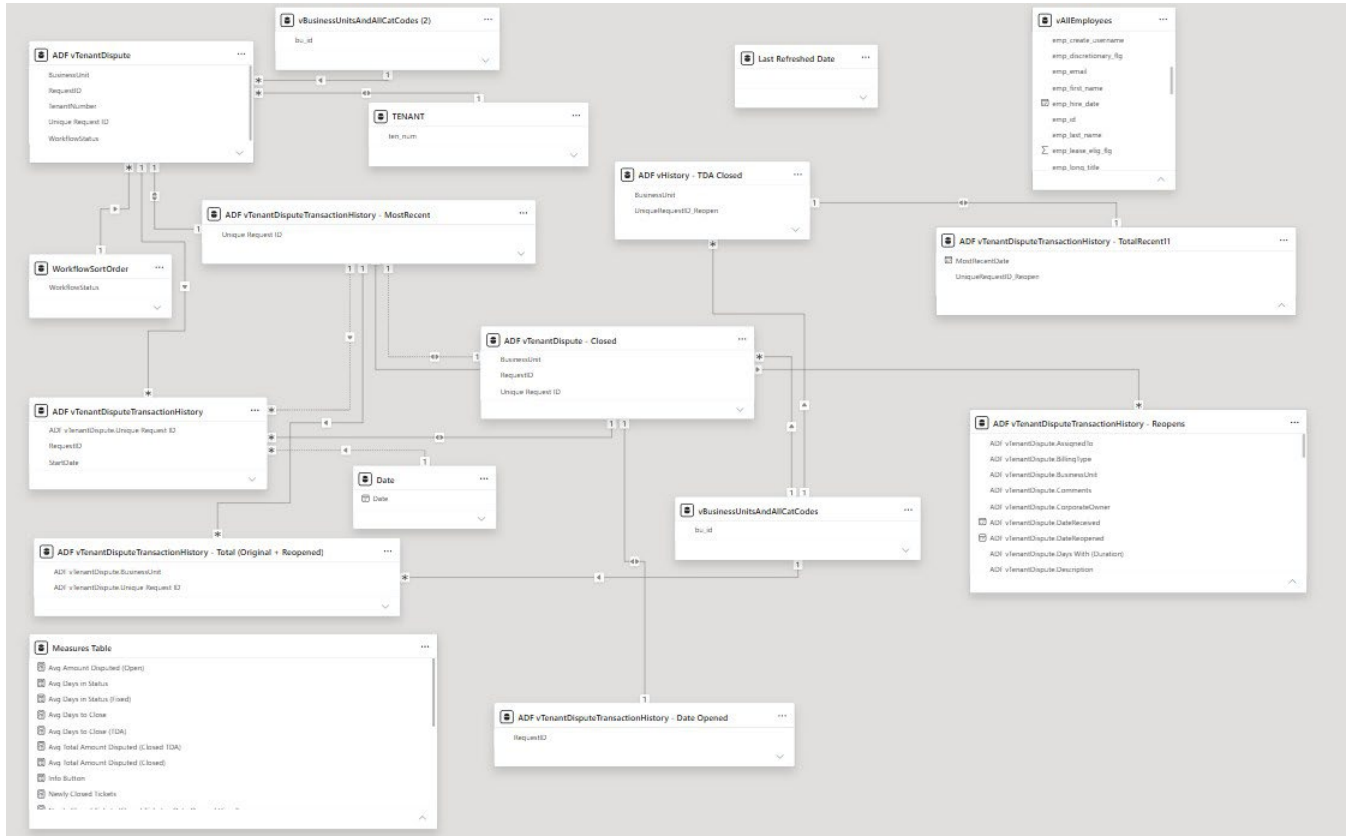
```
SELECT [TotalAmountDisputed], [AssignedTo], [TenantAccountSpecialist], [TotalAdjustmentAmount], [IssueType], [Comments], [FeedbackToTrust], [ModifiedDate], [WorkFlowStatus], [LFKEY_TRACKMASTER] FROM [ADF].[vTenantDispute_V2] ORDER BY [ModifiedDate] desc
```

ID	RequestID	PKey	BusinessUnit	EPPropertyAccountant	EPPropertyAcctMgr	GLPropertyAccountant	DivisionCC23	RegionalOfficerCC21	RegionalPropertyManagerCC18	VintageCategoryCC17	VintageCC16
1	4381	5165	5229				West and Southwest			Same Property	Same Property - Regency
2	4380	5164	5228				West and Southwest			Same Property	Same Property - Regency
3	4378	5163	5227				West and Southwest			Same Property	Same Property - Regency
4	3817	4583	4634				Northeast and MidAtlantic			Same Property	Same Property - Regency
5	4379	5162	5226				West and Southwest			Same Property	Same Property - Regency
6	4312	5100	5163				Northeast and MidAtlantic			Same Property	Same Property - Regency
7	4742	4923	4979				West and Southwest			Same Property	Same Property - Regency
8	4085	5068	5130				Southeast and Midwest			Same Property	Same Property - Regency
9	4351	5132	5196				Southeast and Midwest			Same Property	Same Property - Regency
10	4322	5106	5169				West and Southwest			Same Property	Same Property - Regency
11	4327	5105	5168				West and Southwest			Same Property	Same Property - Regency
12	4328	5104	5167				West and Southwest			Same Property	Same Property - Regency
13	4375	5159	5223				West and Southwest			Same Property	Same Property - Regency
14	4374	5158	5222				West and Southwest			Same Property	Same Property - Regency
15	4377	5161	5225				Southeast and Midwest			Same Property	Same Property - Regency
16	4489	4676	4731				West and Southwest			Same Property	Same Property - Regency

SQL: vTenantDispute_V2 view with PAM, PM, and officer data

Data Model: Semantic Model Architecture

14-table semantic model built from scratch on data warehouse views. Key design elements: separate tables for transaction history, most-recent records, reopens, and a WorkflowSortOrder table for custom status ordering in visuals.



14-table semantic model: transaction history, reopens, most-recent records, employee lookup, and custom workflow sort

3. Tax Invoice Email Automation (Ryan LLC)

125+ hrs	Unattended	v2	Auto-alert
Saved annually	Fully automated	In production	On unroutable files

THE PROBLEM

The property tax vendor pays state, county, and city property taxes on behalf of the company and sends copies of all invoices (both Tax Savings invoices for their services and the underlying Tax Collector invoices). A Property Accounting Administrator manually opened each PDF and filed it to the correct Business Unit folder. As different states' tax due dates created invoice bursts throughout the year, this became a significant and repetitive time drain.

THE SOLUTION

- PowerShell script monitors a shared mailbox for incoming vendor emails. This is running automatically via Power Automate
- Utilizes the vendor's API excel export, Business Unit Numbers at the front of PDFs, and/or parcel mapping to categorize and map each invoice
- Analyzes email subject lines to classify each invoice as a Tax Collector invoice or Tax Savings invoice and routes to separate folder structures
- Timestamped action and error logs written for every run; unroutable invoices trigger automatic email alert to admin and manager
- Runs fully unattended. Version 2 in production with expanded routing logic and improved mapping coverage

THE RESULT

- 125+ hours saved annually, eliminating a fully manual, repetitive PDF filing process
- Admin freed to take on new responsibilities including supporting the Tenant Disputes Manager
- Version 2 live in production with improved parcel mapping and multi-type invoice handling

This automation demonstrates production-grade scripting: configuration section, multi-directory setup, timestamped logging, error handling, and automatic escalation email.

Production Code — PowerShell Automation Script (v2)

Screenshot of production script showing: configuration variables (mailbox, folder paths, mapping files), directory initialization and logging functions (Write-Log and Write-Err). Configuration section at the top allows easy annual updates when folder paths change.

```
File Edit Selection View Go Run Terminal Help
RyanTaxInvoiceAutomation_Prodv2.ps1 2 x Release Notes: 1.121.0
C: > Users > keithnimitz > OneDrive - [redacted] > Ryan Tax Invoices Automation Project > RyanTaxInvoiceAutomation_Prodv2.ps1 > ...

1 # =====
2 # Ryan Tax Invoice Automation
3 # =====
4
5
6 # =====
7 # CONFIGURATION
8 # =====
9
10 $MailboxName = "Tax Savings"
11 $InboxFolderName = "Inbox"
12
13 # Required working directory structure:
14 # *****NEED TO UPDATE GOING INTO NEXT YEAR*****
15 # [redacted]\Reconciliation Letters\2026 Reconciliation Letters\_Tax Invoices\Working\Email_YYYYMMDD_HHMMSS
16 $WorkingRoot = "[redacted]\Reconciliation Letters\2026 Reconciliation Letters\_Tax Invoices\Working"
17
18 #*****NEED TO UPDATE GOING INTO NEXT YEAR*****
19 # Correct routing outputs:
20 $TaxDueRoot = "[redacted]\Reconciliation Letters\2026 Reconciliation Letters\_Tax Invoices"
21 $NonTaxRoot = "[redacted]\Reconciliation Letters\2026 Reconciliation Letters\_RET Supplemental (Tax Savings-Refunds)"
22 $ErrorRoot = "[redacted]\Reconciliation Letters\2026 Reconciliation Letters\_EmailAutomationErrors"
23
24 # Mapping files
25 $ParcelMapping = "[redacted]\Reconciliation Letters\2026 Reconciliation Letters\_EmailAutomationErrors\Parcel BU Mapping.xlsx"
26 $BUMappingFile = "[redacted]\Reconciliation Letters\2026 Reconciliation Letters\_EmailAutomationErrors\BusinessUnitMapping.xlsx"
27
28 # Logs
29 $MainLog = "[redacted]\Reconciliation Letters\2026 Reconciliation Letters\_EmailAutomationErrors\MainLog_Prod.txt"
30 $ErrorLog = "[redacted]\Reconciliation Letters\2026 Reconciliation Letters\_EmailAutomationErrors\ErrorLog_Prod.txt"
31
32
33 # =====
34 # INITIAL SETUP
35 # =====
36
37 foreach ($p in @($WorkingRoot,$TaxDueRoot,$NonTaxRoot,$ErrorRoot)) {
38 | if (!(Test-Path $p)) { New-Item $p -ItemType Directory | Out-Null }
39 }
40
41 if (!(Test-Path $MainLog)) { New-Item $MainLog -ItemType File | Out-Null }
42 if (!(Test-Path $ErrorLog)) { New-Item $ErrorLog -ItemType File | Out-Null }
43
44 19 references
45 Function Write-Log($m) { Add-Content $MainLog "$($Get-Date -Format 'yyyy-MM-dd HH:mm:ss') $m" }
46 16 references
47 Function Write-Err($m) { Add-Content $ErrorLog "$($Get-Date -Format 'yyyy-MM-dd HH:mm:ss') $m" }
```

PowerShell v2: Production script with configuration, directory setup, error logging, and automated routing logic

4. Security Deposit Reconciliation Dashboard

\$250K+	12,500+	2 views
Deposit recovery (first weeks)	Active leases tracked	Not Received + All SECDs

THE PROBLEM

There had never been a portfolio-wide way to identify which tenants had not paid their security deposits, or to reconcile differences between what was held versus what was required per lease. If a tenant does not pay their security deposit it leaves the company exposed to potential losses from damaged units, tenant defaults, and vacated spaces.

THE SOLUTION

- Built a 12-table Power BI semantic model merging JDE ERP security deposit data with ReMaap lease annotation data
- Conducted stakeholder conversations across teams to understand how data flows between the ERP, ReMaap, and the VersaPay payment integration - accounting for edge cases before building
- 'SECDs Not Received' view identifies all active tenants with no deposit on record
- 'All SECDs' view shows every active tenant's deposit balance versus their lease requirement, flagging differences and direction
- Built-in FAQ page educating cross-functional users on how to interpret the data and who to contact by issue type
- Figures displayed are filtered to a subset for portfolio anonymization

THE RESULT

- \$250K+ reduction in unpaid security deposit balances within the first weeks of deployment
- First-ever portfolio-wide view of security deposit compliance — enabling systematic follow-up that was not previously possible
- Cross-functional adoption by Property Management, Property Operations, and Leasing teams

Key design challenge: security deposit data flows through three systems (JDE ERP, ReMaap, VersaPay) with different identifiers and timing. Joining them accurately required stakeholder conversations and careful validation before the report could be trusted.

Dashboard View 1: SECDs Not Received

Identifies active tenants who have not paid their security deposit at all. Filterable by Division, Region, Shopping Center, Regional Property Manager, Property Manager, and GL PAM. Toggle by T Has Options and Lease Expiration timeline.

Security Deposits
Power BI

SECDs Not Received

All SECDs (Required vs Open)

Division, Region, Shopping Center

Regional Property Manager, Property Manager

GL PAM, GL PA

T Has Options

Lease Expiration

Business Unit	Lease	Tenant Name	Deposit Description	Amount per Lease	SECD Balance	Difference per Lease vs Balance	Beg Effective Date	End Effective Date	T Has Options
			Security Deposit	300.00		300.00	4/14/2025	7/31/2026	N
			Security Deposit	300.00		300.00	7/1/2025	7/31/2026	N
			Security Deposit	600.00		600.00	8/2/2023	8/31/2026	N
			Security Deposit	4,144.83		4,144.83	10/1/1996	9/30/2026	N
			Security Deposit	4,424.12		4,424.12	9/18/2020	9/30/2026	N
			Security Deposit	2,300.00		2,300.00	5/1/2026	10/31/2026	N
			Security Deposit	11,153.62		11,153.62	4/1/2024	10/31/2026	N
			Security Deposit	300.00		300.00	8/1/2025	11/30/2026	N
			Security Deposit	6,131.08		6,131.08	12/10/2009	11/30/2026	N
			Security Deposit	1,950.00		1,950.00	1/1/2026	12/31/2026	N
			Security Deposit	7,087.79		7,087.79	12/15/2016	12/31/2026	N
			Security Deposit	9,956.67		9,956.67	12/15/2019	12/31/2026	N
			Security Deposit	17,000.00		17,000.00	3/1/1996	12/31/2026	N
			Security Deposit	23,387.00		23,387.00	1/1/2009	12/31/2026	Y
			Security Deposit	1,000.00		1,000.00	1/1/2026	12/31/2026	N
			Security Deposit	1,400.00		1,400.00	1/1/2026	12/31/2026	N
			Security Deposit	6,696.22		6,696.22	12/22/2006	12/31/2026	Y
			Security Deposit	3,109.75		3,109.75	2/1/2016	1/31/2027	N
			Security Deposit	8,800.90		8,800.90	9/20/2021	2/28/2027	Y
			Security Deposit	12,715.00		12,715.00	2/4/2007	2/28/2027	N
			Security Deposit	2,900.00		2,900.00	2/20/2025	2/28/2027	N
			Security Deposit	2,495.00		2,495.00	3/1/2026	3/1/2027	N
			Security Deposit	2,495.00		2,495.00	4/1/2026	3/31/2027	N
			Security Deposit	4,353.09		4,353.09	11/1/2002	3/31/2027	N
			Security Deposit	3,150.00		3,150.00	4/1/2017	3/31/2027	Y
Total			Security Deposit	2,400.00		2,400.00	4/15/2022	2/23/2027	N

SECDs Not Received: Active tenants with no security deposit on record; subset shown for confidentiality

Dashboard View 2: All SECDs: Lease Requirement vs Balance

All active security deposits showing Amount per Lease, SECD Balance on hand, and the difference between the two.

Security Deposits
Power BI

NOTE: A difference **DOES NOT** necessarily mean a Tenant owes money. See FAQ's for more info.

SECDs Not Received

All SECDs (Required vs Open)

Division, Region, Shopping Center
All

Regional Property Manager, Property Manager
All

GL PAM, GL PA
All

T Has Options
☐ Y
☐ N

Lease Expiration
☐ > 12 Months
☐ Expiring within 12 Months
☐ Expiring within 6 Months

All Active Security Deposits - Amount per Lease vs Amount Open

Business Unit	Lease	Tenant Name	Deposit Description	Amount per Lease	SECD Balance	Difference per Lease & Balance	Beg Effective Date	End Effective Date	T Has Options
			Security Deposit	5,652.76	5,652.76	0.00	6/18/2021	6/30/2026	Y
			Security Deposit	2,478.67	2,478.67	0.00	7/1/2022	6/30/2026	N
			Security Deposit	4,374.07	4,374.07	0.00	3/1/1994	6/30/2026	N
			Security Deposit	31,184.79	31,184.79	0.00	9/25/2006	6/30/2026	N
			Security Deposit	9,315.79	9,315.79	0.00	6/24/2021	6/30/2026	Y
			Security Deposit	1,386.66	1,386.66	0.00	7/1/2016	6/30/2026	N
			Security Deposit	12,144.28	12,144.28	0.00	6/2/2016	6/30/2026	Y
			Security Deposit	1,450.00	1,450.00	0.00	6/15/2020	6/30/2026	N
			Security Deposit	9,300.00	9,300.00	0.00	6/1/2012	6/30/2026	N
			Security Deposit	300.00	300.00	0.00	3/15/2023	6/30/2026	N
			Security Deposit	300.00	300.00	0.00	10/1/2023	6/30/2026	N
			Security Deposit	17,300.34	17,300.34	0.00	6/8/2001	6/30/2026	N
			Security Deposit	3,000.00	3,000.00	0.00	10/1/2008	6/30/2026	N
			Security Deposit	1,645.20	1,645.20	0.00	12/31/1988	6/30/2026	N
			Security Deposit	3,939.00	3,939.00	0.00	12/12/2019	6/30/2026	N
			Security Deposit	4,894.98	4,894.98	0.00	6/26/2016	6/30/2026	Y
			Security Deposit	7,600.00	7,600.00	0.00	10/25/2025	6/30/2026	N
			Security Deposit	9,544.02	9,544.02	0.00	6/26/2016	6/30/2026	N
			Security Deposit	3,279.40	3,279.40	0.00	6/17/2024	6/30/2026	N
			Security Deposit	2,350.00	2,350.00	0.00	7/1/2022	6/30/2026	N
			Security Deposit	30,888.15	30,888.15	0.00	2/4/2020	6/30/2026	N
			Security Deposit	4,359.80	4,359.80	0.00	6/6/2016	6/30/2026	Y
			Security Deposit	8,652.06	8,652.06	0.00	6/15/2023	6/30/2026	Y
			Security Deposit	25,521.82	25,521.82	0.00	10/7/2009	6/30/2026	N
			Security Deposit	6,825.60	6,825.60	0.00	3/1/1989	6/30/2026	N
Total									

All SECDs: Per-lease required amount vs actual balance with difference analysis; subset shown for confidentiality

Dashboard View 3: FAQ & Resources Page

Built-in documentation page explaining what the dashboard shows, how to interpret differences, and who to contact for each question type (Lease Admin, EP Property Accounting, Accounts Receivable, POAC committee). Includes links to a PDF guide and video introduction for self-service adoption.

The screenshot shows a web browser displaying the 'Security Deposits Power BI - Frequently Asked Questions (FAQ's)' page. The page has a dark header with the title 'Security Deposits Reconciliation Power BI' and a search bar. A left sidebar contains a 'Pages' menu with 'SEC'D Deposits Power BI' and 'FAQ's & Resources'. The main content area is titled 'Security Deposits Power BI - Frequently Asked Questions (FAQ's)' and contains several sections of text and links. The sections include: 'What is the Security Deposits Power BI?', 'How often is the Security Deposits Power BI Refreshed?', 'Where can I find more information on the Security Deposits Power BI?', 'What is the difference between the "SEC'Ds Not Received" view and the "All SEC'Ds (Required vs Open)" view?', 'What does the "Difference per Lease & Balance" mean?', 'Who do I reach out to if I have questions?', 'Where are Letters of Credit - are they included here?', and 'Where are the Ballard properties? I do not see them in this dashboard.' Each section provides detailed explanations and links to relevant resources.

Security Deposits Power BI - Frequently Asked Questions (FAQ's)

What is the Security Deposits Power BI?
The Security Deposits Power BI is a Power BI that summarizes the Security Deposit balances across the portfolio of active tenants. In particular, it helps highlight tenants whose Security Deposits we have not received or where we have differences in our SEC'D Balance and what is required per the lease.

How often is the Security Deposits Power BI Refreshed?
The Security Deposits Power BI is refreshed on a weekly basis.

Where can I find more information on the Security Deposits Power BI?
Additional information and resources can be found in the resource documents and training video linked below:

- [Security Deposits Reconciliation Power BI - PDF Guide.pdf](#)
- [Security Deposits Power BI - Video Introduction.mp4](#)

What is the difference between the "SEC'Ds Not Received" view and the "All SEC'Ds (Required vs Open)" view?
The "SEC'Ds Not Received" shows which active tenants we have not received Security Deposits from yet.

The "All SEC'Ds (Required vs Open)" view shows the SEC'D Balances and what are required per the lease for each active tenant.

What does the "Difference per Lease & Balance" mean?
The "Difference per Lease & Balance" means that there is a difference in the amount required per the lease and the Security Deposit balance related to the tenant.

- If the "SEC'D Balance" appears blank, that means we never received a Security Deposit from the tenant.
- If the difference is 0.00, that means that the tenant's SEC'D Balance agrees with what is required per the lease.
- If the difference is greater than zero, then that means the SEC'D Balance is short from the amount required per the lease. This could be for various reasons and clauses such as required partial refunds, partial applications of the SEC'D, etc. Further research would be needed to determine if the tenant truly owes additional funds.
- If the difference is less than zero, then that means the SEC'D Balance is greater than the amount required per the lease. Similar to the difference being greater than zero, this could be due to various reasons and clauses and would need further research to determine if the tenant truly owes additional funds.

Who do I reach out to if I have questions?
If you have questions please see below for the appropriate contact:

- If you have a question on the "Amount per Lease", please reach out to your Lease Administrator.
- If you have a question on a Security Deposit refund, please reach out to your EP Property Accounting Manager.
- If you have a question on the tool or feedback, please reach out to a member of the Property Ops & Property Accounting Committee
- If you have a question on a Security Deposit application or partial application, please reach out to Accounts Receivable.

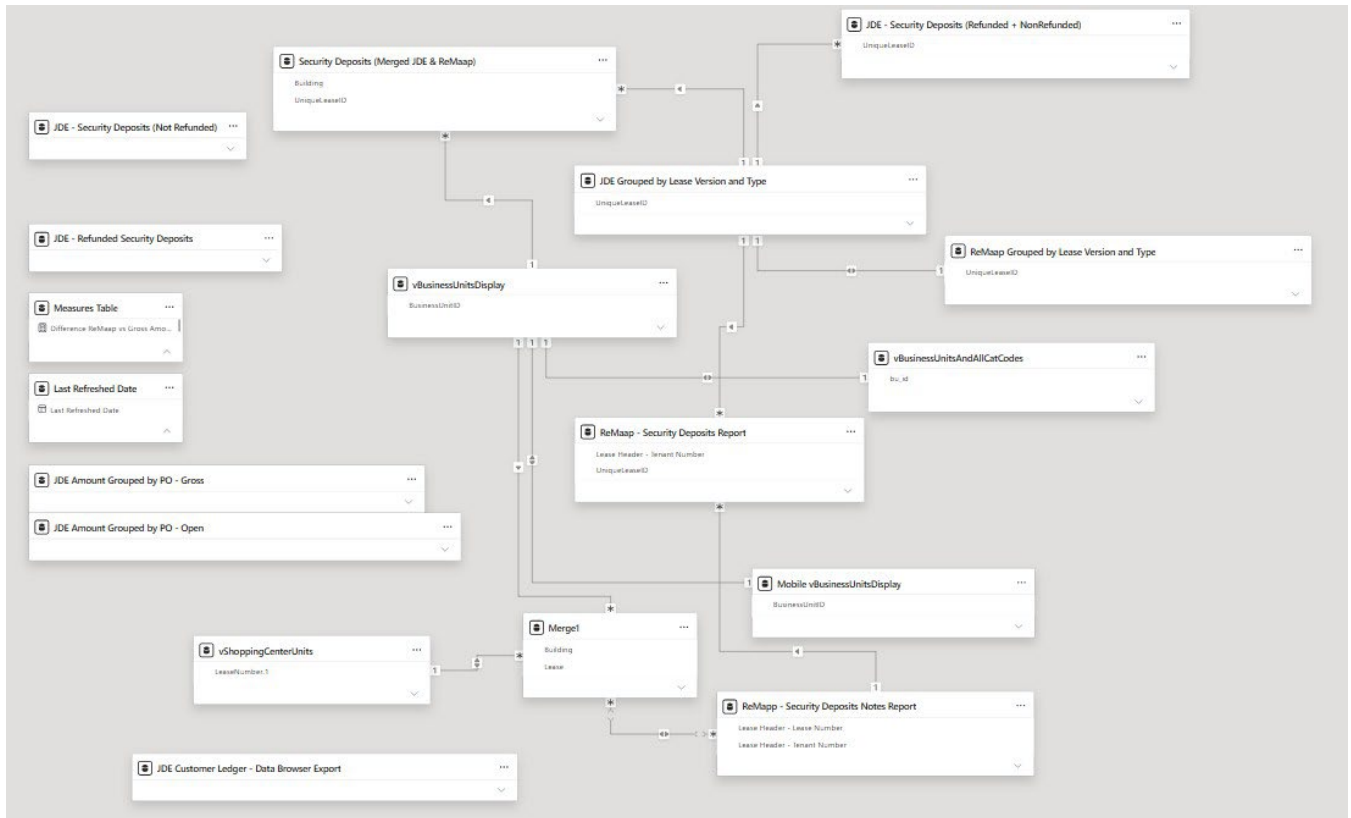
Where are Letters of Credit - are they included here?
No. Letters of Credit are not included in this Power BI.

Where are the Ballard properties? I do not see them in this dashboard.
The Ballard properties are excluded from this dashboard and are not in scope herein.

FAQ page: User documentation, interpretation guide, and contact routing by issue type built directly into the dashboard

Data Model: Semantic Model Architecture

12-table semantic model. Separate tables handle refunded vs non-refunded deposits, grouped lease version data from both systems, and a custom Merge table for the reconciled view.



12-table semantic model: JDE ERP and ReMaap joined via a custom UniqueLease ID field

5. Property Accounting KPI Dashboard

6 metrics	17 tables	YoY
On one dashboard	Semantic model depth	Year-over-year comparison

THE PROBLEM

With re-organizations and changes in the Property Accounting team, management and senior leadership needed a way to see the health, speed, accuracy, and efficiency of their close processes at a high level and to measure whether organizational changes were improving or decreasing these metrics over time. There was no existing tool to answer this question.

THE SOLUTION

- Built a 17-table semantic model connecting JDE Batch History, GL Details, Lease Data, Reclass Data, Time Study Analytics, Closing Dates, and multiple review period tables
- Single-page dashboard with 6 key metric charts: Reclasses Monthly, Net Manual Billings, Post-Review Entries, Percentage Rent Processing, Lease Commissions, and Close Review Timing
- Year-over-year (2025 vs 2026) comparison enabled for all metrics
- Drillthrough pages allow management to click any data point and see the underlying detail
- Filterable by Division, Region, Property Manager, GL PAM, EP PAM, and Review Period

THE RESULT

- Management and senior leadership now have visibility over close health metrics for Quarterly Business Reviews and ongoing performance monitoring
- Time Study Analytics embedded directly enable data-driven resource planning and process improvement tracking over time

The PA KPI Dashboard was built in response to a direct management request, translating a business question ('are our changes working?') into a measurable analytical tool with year-over-year comparison.

Dashboard: Summary Page

Six-metric management dashboard showing Reclasses Monthly, Net Manual Adjustment Billings, Entries Posted After Manager Review, Percentage Rent Processed, Lease Commissions, and entries by review period - all with YoY comparison.



KPI Dashboard Summary: 6 close health metrics with YoY comparison, filterable by division, region, PM, and review period

Dashboard: Drillthrough Example

Drillthrough page accessible from any chart on the summary page. Provides underlying detail records for the selected metric and period, allowing management to move from the summary view to individual journal entry or billing activity level in one click.

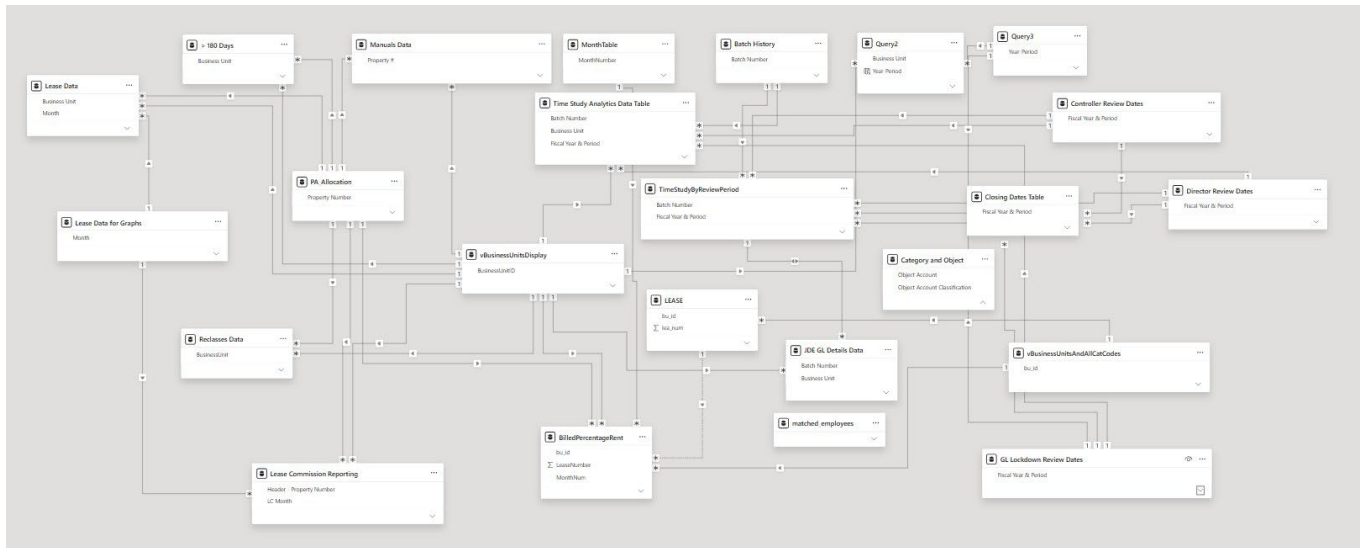


Month	Year	Property Number	Shopping Center Name	Lease #	Tenant #	Tenant Name	Unit Number	Billed Amount
December	2025						1018	\$152,028
December	2025						501	\$128,113
December	2025						8255	\$76,394
December	2025						0009	\$67,072
December	2025						550	\$54,087
December	2025						575	\$53,726
December	2025						153-D	\$45,654
December	2025						11A	\$44,857
December	2025						50	\$37,068
December	2025						A0173	\$36,116
December	2025						58	\$35,814
December	2025						37	\$35,100
December	2025						760	\$31,448
December	2025						0010	\$29,852
December	2025						9812F	\$28,643
December	2025						401	\$27,176
December	2025						1471	\$26,602
December	2025						440	\$24,740
December	2025						J717	\$20,676
December	2025						E0524	\$20,357
December	2025						140	\$17,702
December	2025						1135-E	\$15,354
December	2025						G662	\$14,262
December	2025						13	\$14,220
December	2025						B0218C	\$13,956
December	2025						A0166	\$13,841
December	2025						112	\$13,319
December	2025						20	\$13,384
December	2025						0019	\$12,267
December	2025						560	\$8,861
December	2025						D310	\$8,770
December	2025						K804	\$8,601
December	2025						8	\$8,424
December	2025						738	\$8,272
December	2025						0046	\$8,158
December	2025						J745	\$7,968
December	2025						00001	\$7,377
Total								\$1,443,443

Drillthrough: Underlying detail accessible from any KPI chart with full record-level data

Data Model: Semantic Model Architecture

17-table semantic model. Spans Batch History, GL Detail Data, Lease Data (two versions for graph and tabular views), Reclass Data, Billed Percentage Rent, Lease Commission Reporting, Time Study Analytics, PA Allocation, and four separate review period date tables for precise period-over-period calculations.



17-table semantic model; spanning close analytics, time study, and multiple review period dimensions

6. Lease Caps & Exclusions (Power BI & Property Connect Integration)

12,500+	CRM
Tenant leases covered	Integrated in Property Connect

THE PROBLEM

When renegotiating leases for tenants with leases about to renew or expire, it was difficult for Leasing and Property Management teams to identify what lease caps, market caps, account exclusions, and other NOI-limiting clauses were in the lease. Teams entered renewal negotiations without visibility into which clauses could be renegotiated, limiting the ability to improve NOI as leases turned over.

THE SOLUTION

- Built a 15-table Power BI semantic model
- Dashboard shows for each tenant: CAM rec type, management fee exclusions, account exclusions, admin fee exclusions, lease cap presence, cap type (i.e. 5% over PY), and prior-year market and lease cap bleed by component
- Annual payers and cap bleed views provide additional analytical layers for targeting high-value renegotiation opportunities
- Integrated into Property Connect (the company's field-facing CRM) so teams see lease cap data directly in the tenant details view without opening Power BI

THE RESULT

- Leasing and Property Management teams now enter renewal negotiations with full lease cap and exclusion context
- Property Connect integration means zero additional login or system switching — data surfaces where field teams already work daily
- Enables targeted renegotiation strategy for NOI-limiting clauses as leases approach expiration

Key differentiator: This is not just a Power BI dashboard. The data was surfaced inside Property Connect (the company's operational CRM used by field teams).

Dashboard View 1: Main Opening Page

Full tenant roster showing lease status, CAM tenant type, lease options, management fee exclusions, account exclusions, admin fee exclusions, lease cap presence, and prior-year bleed amounts. Filterable by Division, Region, Shopping Center, Regional Officer, Leasing Agent, and Property Manager.

Lease Caps & Exclusions

*\$ Bleed Based on 2024 Reconciliations

Lease Status:

Active

Pre-Lease

Month to Month

Hold-over

T Has Options

☐ Y

☐ N

Lease Expiration

☐ >12 Months

☐ Expiring within 12 Months

☐ Expiring within 6 Months

Division, Region

Shopping Center Name

All

Regional Officer

All

Leasing Agent

All

Regional Prop Manager

All

Property Name	Lease #	Tenant #	Tenant Name	CAM Tenant Type	Lease End Date	T Has Option	T Excludes MGMT Fee	T Has Acct Exclusions	T Excludes Admin Fee	T Has Lease Cap	2024 T Has MRKT Cap	2024 Lease Cap \$ Bleed	2024 MRKT Cap \$ Bleed
				FPR		N	N	N	N	Y	N	0	0
				Other		N	N	N	Y	N	N	0	0
				After Anchor		Y	N	N	N	Y	N	0	0
				After Anchor		N	N	Y	Y	Y	N	8,231	0
				FPR		Y	Y	Y	N	Y	N	1,170	0
				FPR		Y	N	N	Y	N	N	0	0
				After Anchor		N	N	Y	Y	Y	N	0	0
				After Anchor		N	N	Y	Y	Y	N	0	0
				Other		N	N	N	Y	N	N	0	0
				FPR		N	Y	Y	N	N	N	0	0
				After Anchor		N	N	N	Y	N	N	0	0
				FPR		Y	N	Y	N	N	N	0	0
				FPR		N	Y	N	N	N	N	0	0
				Other		N	N	N	Y	N	N	0	0
				After Anchor		N	N	N	Y	N	N	0	0
				After Anchor		Y	N	Y	N	Y	N	4,513	0
				FPR		N	Y	Y	N	Y	N	1,740	0
				After Anchor		N	N	N	Y	N	N	0	0
				After Anchor		N	N	N	N	N	Y	0	5,336
				FPR		N	N	N	Y	N	N	0	0
				FPR		N	N	N	N	N	N	0	0
				Other		N	N	N	Y	N	N	0	0
				FPR		N	N	N	N	N	N	0	0
				Other		N	N	N	Y	N	N	0	0
				Other		N	N	N	Y	N	N	0	0
				After Anchor		N	N	N	Y	N	N	0	0
				After Anchor		N	Y	Y	N	N	N	0	0
				FPR		N	N	N	Y	N	N	0	0
				Other		N	N	N	Y	N	N	0	0

Lease Caps & Exclusions main view: Full tenant roster with key clause fields and prior-year bleed amounts

Dashboard View 2: Drillthrough - Account Exclusions

Drillthrough page showing the specific account exclusions for a selected tenant and what cost categories are excluded from their CAM reconciliation. Critical context for understanding the true CAM exposure per tenant and identifying renegotiation opportunities.

⬅

Drillthrough - Account Exclusions

Property Name	Lease #	Lease Version	Tenant Name	Anchor/Shop	CAM Recovery Type	CAM Tenant Type	A L	E.P. Lease Adj. Level Type	Obj.Sub	Description [F0901]
				Shop	PRS	After Anchor	A	3	7115.UX	TENANT ELECTRICITY EXTRAP
				Shop	PRS	After Anchor	A	3	7125.UX	TENANT WATER SEWER EXTRAP
				Shop	PRS	After Anchor	A	3	7210.UX	ELECTRICITY EXTRAP
				Shop	PRS	After Anchor	A	3	7240.UX	IRRIGATION SEWER EXTRAP
				Shop	PRS	After Anchor	A	3	7320.UX	WASTE LITTER REMOVAL EXTRAP
				Shop	PRS	After Anchor	A	3	7435.UX	LIFE SAFETY EXTRAP

Drillthrough: Account Exclusions showing specific CAM exclusion categories per tenant

Dashboard View 3: Drillthrough - Lease & Market Cap \$ Bleed

Shows the dollar bleed from lease and market caps for a selected tenant and how much recovery was limited by the caps in the prior year. The most actionable view for lease renegotiation, identifying which tenants represent the highest NOI recovery opportunity.

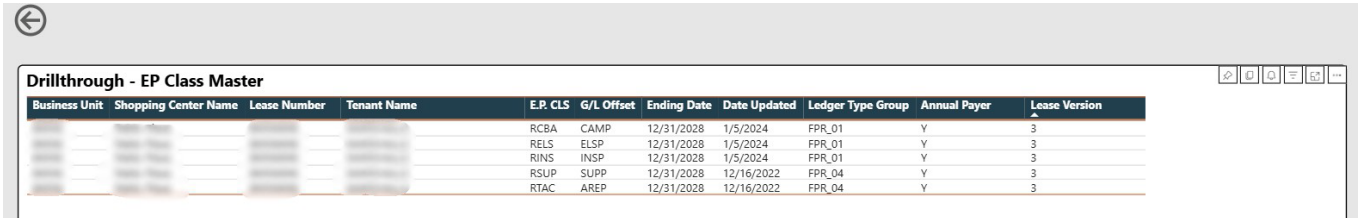
←

Lease #	Tenant Name	Anchor/ Shop	CAM Recovery Type	CAM Tenant Type	CAM Lease Cap \$ Bleed	Cap Rule - CAM	INS Lease Cap \$ Bleed	Cap Rule - INS	RET Lease Cap \$ Bleed	Cap Rule - RET	2024 Lease Cap \$ Bleed	CAM MRKT Cap \$ Bleed	INS MRKT Cap \$ Bleed	RET MRKT Cap \$ Bleed	2024 MRKT Cap \$ Bleed
		Shop	PRS	After Anchor	2,740	5%PY	0	N	0	N	2,740	0	1,859	0	1,859
Total					2,740		0		0		2,740	0	1,859	0	1,859

Drillthrough: Lease & Market Cap bleed by component (CAM, INS, TAX) based on prior year reconciliations

Dashboard View 4: Drillthrough - EP Class Master

EP Class Master drillthrough showing lease class codes, GL offsets, lease ending dates, annual payer status, and lease version for the selected tenant. Provides the EP Property Accounting context needed to understand how the tenant's charges are structured.



Business Unit	Shopping Center Name	Lease Number	Tenant Name	E.P. CLS	G/L Offset	Ending Date	Date Updated	Ledger Type Group	Annual Payer	Lease Version
				RCBA	CAMP	12/31/2028	1/5/2024	FPR_01	Y	3
				RELS	ELSP	12/31/2028	1/5/2024	FPR_01	Y	3
				RINS	INSP	12/31/2028	1/5/2024	FPR_01	Y	3
				RSUP	SUPP	12/31/2028	12/16/2022	FPR_04	Y	3
				RTAC	AREP	12/31/2028	12/16/2022	FPR_04	Y	3

Drillthrough: EP Class Master with class codes, GL offsets, and annual payer designation

Dashboard View 5: Annual Payers

Separate view filtering to annual payer tenants only, showing their lease cap status, exclusions, and bleed amounts. Annual payers reconcile once per year rather than monthly, making their cap and exclusion terms particularly important to understand before renegotiation.

Annual Payers

Lease Status:

Active

Pre-Lease

Month to Month

Hold-over

T Has Options

Lease Expiration

Division, Region

Shopping Center Name

Regional Officer

Leasing Agent

Regional Prop Manager

Regional Officer	Property Name	Lease #	Tenant #	Tenant Name	CAM Tenant Type	Lease End Date	T Has Option	Annual Payer?	CAM Annual Payer?	INS Annual Payer?	RET Annual Payer?
					FPR		N	Y	N	Y	Y
					Other		N	Y	N	N	Y
					Other		Y	Y	N	N	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	Y	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	Y	Y
					FPR		Y	Y	N	N	Y
					Other		Y	Y	N	N	Y
					FPR		Y	Y	N	N	Y
					Other		N	Y	N	N	Y
					FPR		Y	Y	N	Y	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	Y	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	Y	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	N	Y
					Other		N	Y	Y	N	N
					After Anchor		Y	Y	N	N	Y
					FPR		Y	Y	N	Y	Y
					FPR		Y	Y	N	Y	Y
					FPR		Y	Y	N	Y	Y
					FPR		N	Y	N	N	Y
					FPR		Y	Y	N	N	Y
					FPR		N	Y	N	Y	Y
					FPR		Y	Y	N	N	Y
					FPR		Y	Y	N	Y	Y
					FPR		Y	Y	N	Y	Y

Annual Payers view: Lease caps and exclusions filtered to annual reconciliation tenants

Dashboard View 6: Lease & Market Cap \$ Bleed (All Tenants)

Portfolio-wide view of lease and market cap bleed showing the total dollar impact across all tenants. Allows management to see the aggregate impact of existing cap structures and prioritize renegotiation efforts by scale of opportunity.

	Lease/Market Cap \$ Bleed														
	*\$ Bleed Based on 2024 Reconciliations														
Division, Region		Shopping Center Name		Regional Officer		Leasing Agent		Regional Prop Manager				T Has Options		Lease Expiration	
		All				All						☐ Y		☐ > 12 Months	
												☐ N		☐ Expiring within 12 Months	
														☐ Expiring within 6 Months	
Lease #	Tenant Name	Anchor/ Shop	CAM Recovery Type	CAM Tenant Type	CAM Lease Cap \$ Bleed	Cap Rule - CAM	INS Lease Cap \$ Bleed	Cap Rule - INS	RET Lease Cap \$ Bleed	Cap Rule - RET	2024 Lease Cap \$ Bleed	CAM MKRT Cap \$ Bleed	INS MKRT Cap \$ Bleed	RET MKRT Cap \$ Bleed	2024 MKRT Cap \$ Bleed
	Shop	PRS	After Anchor		24,937	5% CUM	0	N	0	N	24,937	0	0	0	0
	Anchor	PRS	FPR	0	N		24,068	FIXED	0	N	24,068	0	0	0	0
	Anchor	PRS	FPR		23,265	5% PY	0	N	0	N	23,265	0	0	0	0
	Anchor	PRS	FPR		21,436	5% PY	0	N	0	N	21,436	0	0	0	0
	Shop	PRS	After Anchor		17,457	5% PY	0	N	0	N	17,457	0	0	0	0
	Shop	PRS	After Anchor		15,968	5% PY	0	N	0	N	15,968	0	0	0	0
	Shop	PRS	After Anchor		14,023	5% PYRF	0	N	0	N	14,023	0	0	0	0
	Shop	PRS	After Anchor		13,895	5% PY	0	N	0	N	13,895	0	0	0	0
	Shop	PRS	After Anchor		12,294	5% PY	0	N	0	N	12,294	0	0	0	0
	Anchor	PRS	FPR		0	FIXED	9,257	7.5% CUM	0	N	9,257	0	0	0	0
	Anchor	PRS	FPR		8,547	2% PY	0	N	0	N	8,547	0	0	0	0
	Anchor	PRS	FPR		0	N	7,687	25% CUM	0	N	7,687	0	0	0	0
	Anchor	PRS	FPR		7,610	2% CUM	0	N	0	N	7,610	0	0	0	0
	Anchor	PRS	FPR		6,512	5% CUM	0	7.5% CUM	0	5% CUM	6,512	0	0	0	0
	Shop	PRS	After Anchor		5,361	5% PY	0	N	0	N	5,361	0	0	0	0
	Shop	PRS	FPR		5,156	3% PY	0	N	0	N	5,156	0	0	0	0
	Shop	PRS	FPR		5,147	5% PY	0	N	0	N	5,147	0	0	0	0
	Shop	PRS	After Anchor		5,009	5% CUM	0	N	0	N	5,009	0	0	0	0
	Shop	PRS	After Anchor		4,808	5% PY	0	N	0	N	4,808	0	0	0	0
	Shop	PRS	After Anchor		4,513	5% PY	0	N	0	N	4,513	0	0	0	0
	Shop	PRS	After Anchor		4,419	5% PY	0	N	0	N	4,419	2,314	0	0	2,314
	Anchor	PRS	FPR		4,049	5% PY	0	N	0	N	4,049	0	0	0	0
	Shop	PRS	FPR		3,931	3% PY	0	N	0	N	3,931	0	0	0	0
Total					270,146		41,686		0		311,832	495,480	16,127	5,716	517,323

Lease & Market Cap Bleed: All tenants showing aggregate prior-year bleed by cap type and component

Property Connect Integration: Lease Caps in Tenant Details

The Lease Caps & Exclusions data surfaced directly in Property Connect (the field-facing CRM used by Leasing Agents and Property Managers). Visible in the Tenant Details view without needing to open Power BI. Built in coordination with IT and Operations teams.

Lease Caps & Exclusions - Lease

Power BI

Overview - Property Connect

propertyconnect/Tenants/Overview/0006A/

Lease Caps & Exclusions

CAM Rec Type	T Excludes MGMT Fee	T Excludes Other Accts	T Excludes Admin Fee	T Has Lease Cap	2024 LL Market Cap Bleed (NNN)	
After Anchor	N	N	N	Y	\$2,314.00	
2024 Reconciliation Cap \$						
CAM Cap Rule	INS Cap Rule	TAX Cap Rule	CAM Cap Bleed	INS Cap Bleed	TAX Cap Bleed	Total Lease Cap Bleed
5%PY	N	N	\$4,419.00	\$0.00	\$0.00	\$4,419.00

Last Updated: 10/24/2024 10:00 AM

Property Connect integration: Lease Caps & Exclusions appearing in the Tenant Details view of the field CRM

SQL Query Against Data Warehouse

SQL query of (jde_db_reggie.dbo.LeaseCapsAndExclusions) table. This table is a flat table of the Power BI data that was loaded here to be integrated into Property Connect for this project. Future phases of the project are in progress to make this a true data warehouse view table and import/load the appropriate supporting tables.

The screenshot shows the Microsoft SQL Server Enterprise Manager interface. On the left, the 'Object Explorer' pane displays a tree view of the 'jde_db_reggie' database, with 'dbo.LeaseCapsAndExclusions' selected. The main window displays a SQL query in the 'SQL Query Editor' pane. The query is a SELECT statement with a TOP clause, selecting various columns from the 'dbo.LeaseCapsAndExclusions' table. The results pane at the bottom shows the output of the query, which is a table with 13 columns and 25 rows. The columns are: LeaseNumber, TenantNumber, TenantName, CAM_Rec_Type, T_Excludes_MGMT_Fee, T_Has_Acct_Exclusions, T_Excludes_Admin_Fee, T_Has_Lease_Cap, CAM_Cap_Rule, INS_Cap_Rule, TAX_Cap_Rule, 2024_LL_Market_Cap_Bleed_NNN, and CAM_Cap_Bleed. The results are sorted by LeaseNumber in descending order.

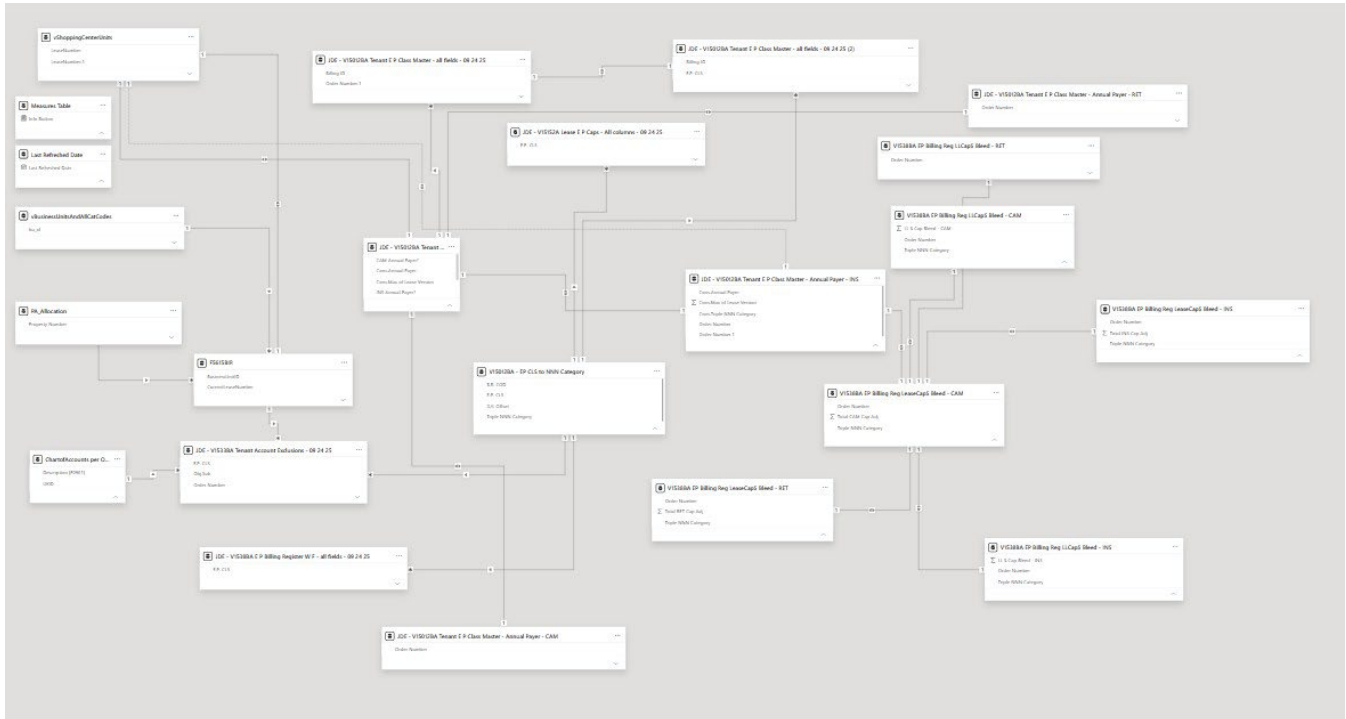
```
SELECT TOP (1000) [LeaseNumber]
,[TenantNumber]
,[TenantName]
,[CAM_Rec_Type]
,[T_Excludes_MGMT_Fee]
,[T_Has_Acct_Exclusions]
,[T_Excludes_Admin_Fee]
,[T_Has_Lease_Cap]
,[CAM_Cap_Rule]
,[INS_Cap_Rule]
,[TAX_Cap_Rule]
,[2024_LL_Market_Cap_Bleed_NNN]
,[CAM_Cap_Bleed]
,[RET_Cap_Bleed]
,[Total_Lease_Cap_Bleed]
,[ModelYear]
,[ImportDate]
FROM [jde_db_reggie].[dbo].[LeaseCapsAndExclusions]
```

LeaseNumber	TenantNumber	TenantName	CAM_Rec_Type	T_Excludes_MGMT_Fee	T_Has_Acct_Exclusions	T_Excludes_Admin_Fee	T_Has_Lease_Cap	CAM_Cap_Rule	INS_Cap_Rule	TAX_Cap_Rule	2024_LL_Market_Cap_Bleed_NNN	CAM_Cap_Bleed
710			After Anchor	N	N	Y	N	N	N	N	0	0
728			FPR	Y	N	Y	N	N	N	N	0	0
762			FPR	Y	Y	N	Y	6 SUPY	N	N	0	0
769			After Anchor	N	N	Y	N	10 SUPY	N	N	0	0
1501			FPR	N	N	Y	N	N	N	N	0	0
1503			FPR	N	N	Y	N	N	N	N	0	0
1511			Other	N	N	Y	N	N	N	N	0	0
1517			FPR	N	N	Y	N	N	N	N	0	0
1525			FPR	N	N	Y	N	N	N	N	0	0
1527			FPR	N	N	Y	N	N	N	N	0	0
1528			After Anchor	N	N	Y	N	N	N	N	0	0
1531			FPR	N	N	Y	N	N	N	N	0	0
1533			FPR	N	Y	Y	N	N	N	N	0	0
1537			Other	N	N	Y	N	N	N	N	0	0
1539			FPR	N	Y	Y	N	N	N	N	0	0
1541			FPR	N	Y	Y	Y	71CUMRF	N	N	0	0
1545			FPR	N	Y	Y	Y	431CUMCPI	N	N	0	72633
1548			FPR	Y	Y	Y	N	N	N	N	0	0
1554			After Anchor	N	Y	Y	N	N	N	N	0	0
1558			After Anchor	Y	N	N	N	N	N	N	0	0
1559			After Anchor	N	N	Y	N	N	N	N	0	0
1563			FPR	Y	N	N	Y	SUPY	N	N	0	0
1565			After Anchor	N	N	Y	N	N	N	N	0	0
1570			After Anchor	N	N	Y	Y	81CUM	N	N	0	0

SQL Query against jde_db_reggie.dbo.LeaseCapsAndExclusions

Data Model: Semantic Model Architecture

15-table semantic model joining JDE tenant master tables (multiple versions for EP class), billing registers, and custom shopping center hierarchy. Key challenge: joining EP and non-EP tenant data with different class master structures while maintaining accurate filtering.



15-table semantic model: JDE tenant master, billing registers, EP class data, and shopping center hierarchy